



MCT333 : Mechatronics System Design

Lecture 02: Design Methodology for Mechatronics system (VDI 2206)

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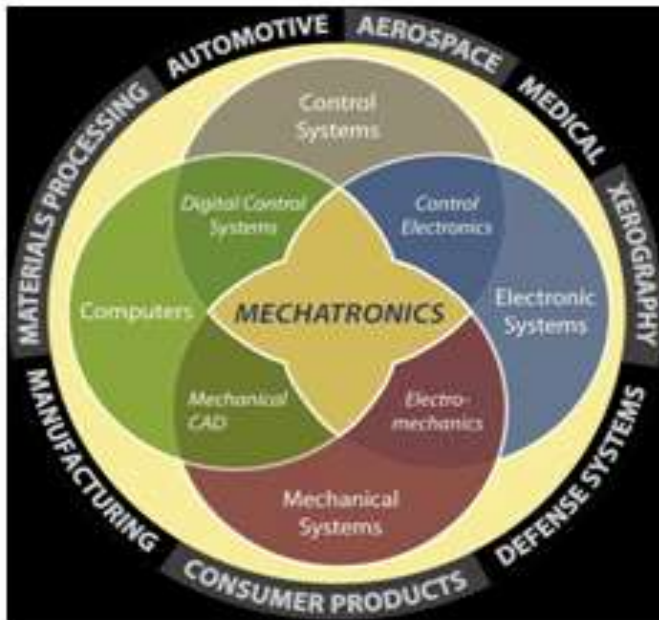
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Definitions

- Mechatronics:

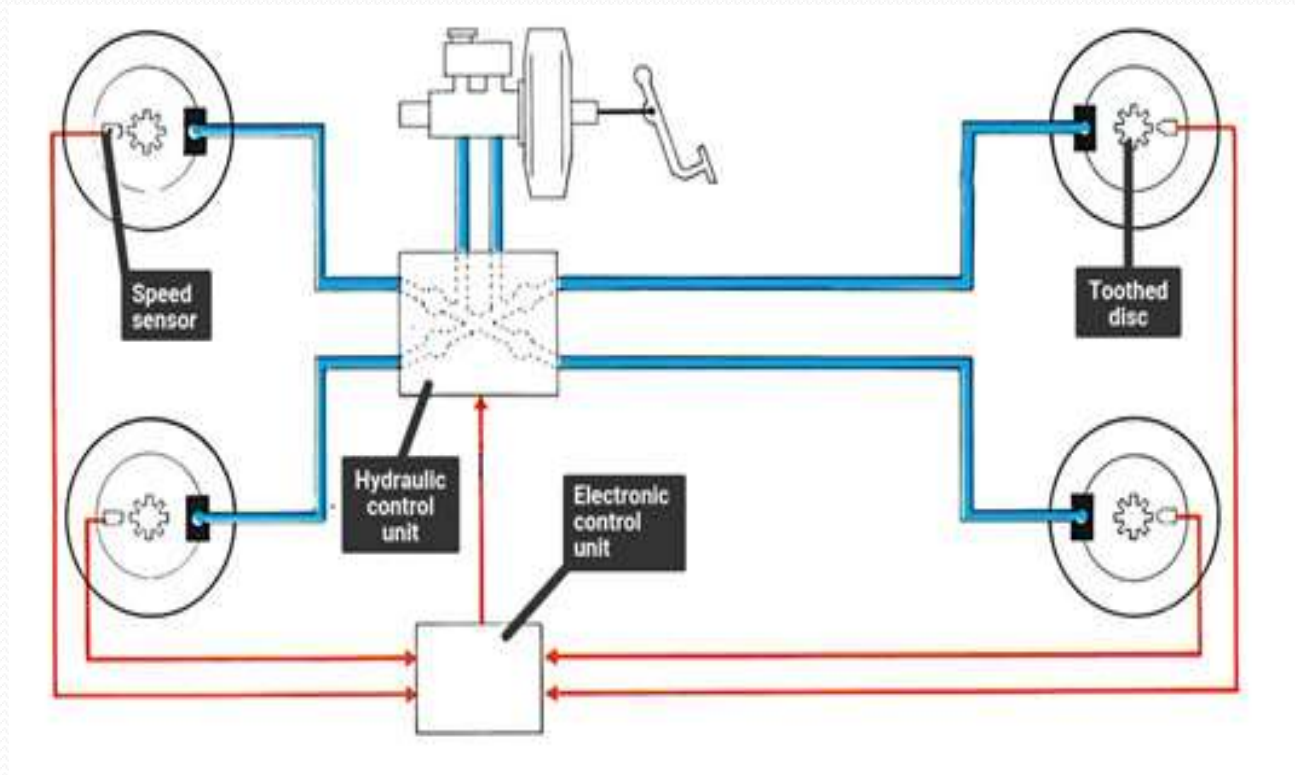


Mechatronics is synergistic integration of mechanical engineering, electronic systems, control systems and intelligent control in design and manufacture of products and processes.

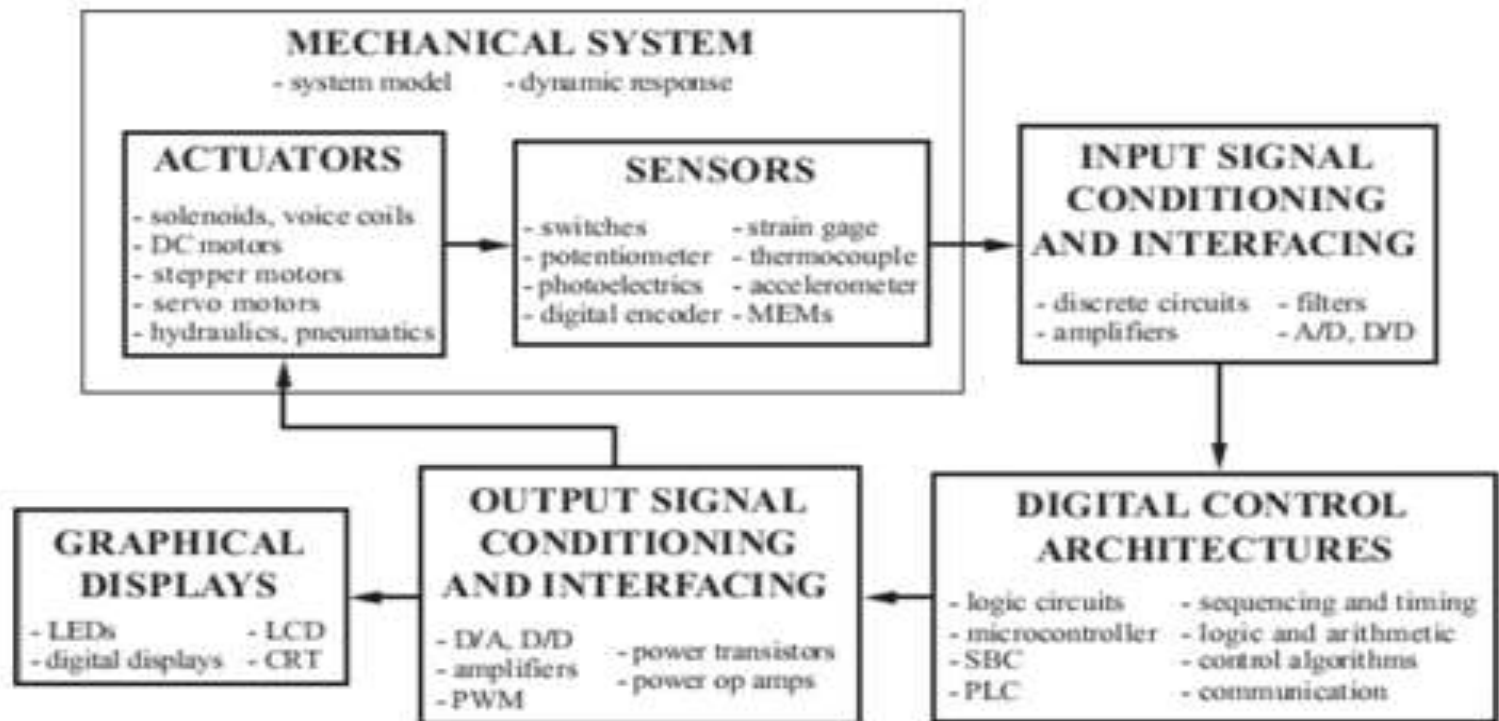
Examples



Examples



Mechatronic Systems Components



Why Mechatronics?

- **Mechatronics is a design philosophy:** an integrating approach to engineering design
- Mechatronics is a **methodology** used for the optimal design of inter-disciplinary products.
- The mechatronic design methodology is based on a **concurrent** (instead of sequential) approach to discipline design, resulting in products with more synergy.

Why Mechatronics?

- Mechatronics is **a synergy in the integration** of mechanical, electrical, and computer systems with information systems for the design and manufacturing of products and processes. The synergy is generated by the right combination of parameters; **the final product can be better than just the sum of its parts.**
- Mechatronic products exhibit performance characteristics that were previously difficult to achieve

Why Mechatronics?

- Concurrent engineering is a design approach in which the design and manufacture of a product are merged in a special way. It is the idea that people can do a better job if they cooperate to achieve a common goal. It has been influenced partly by the recognition that many of the high costs in manufacturing are decided at the product design stage itself.

Why Mechatronics?

1. Better definition of the product without late changes.
2. Design for manufacturing and assembly undertaken in the early design stage.
3. Process on how the product development is well defined.
4. Better cost estimates.
5. Decrease in the barriers between design and manufacturing.



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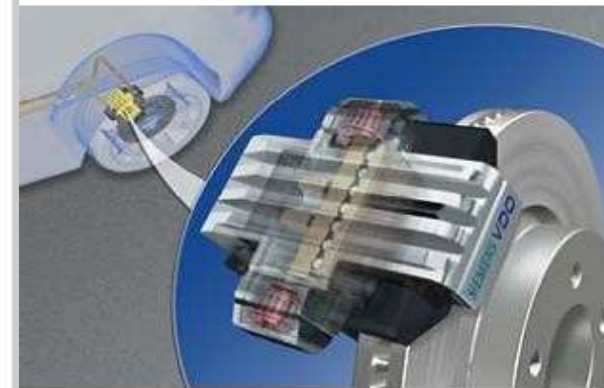
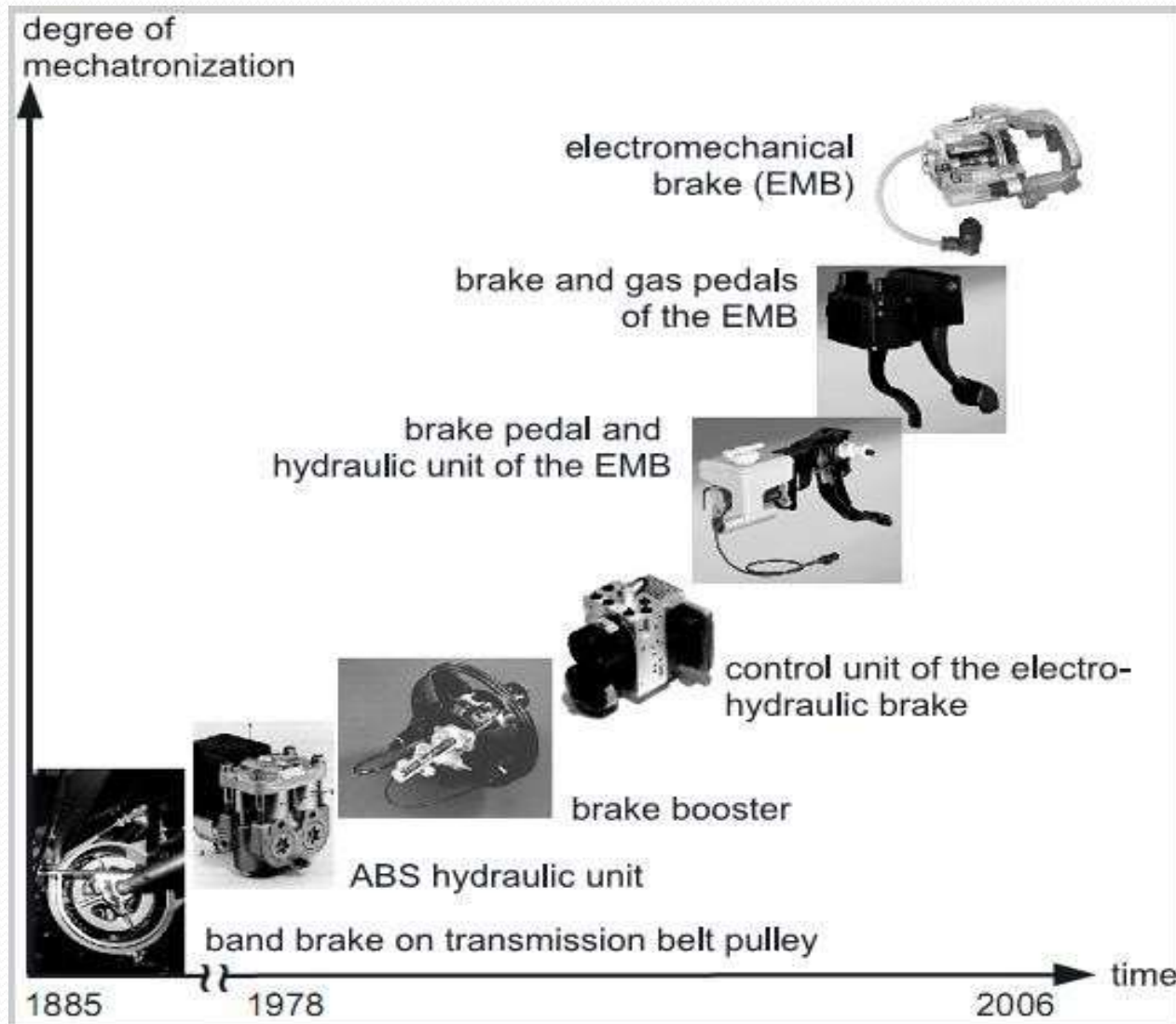
Entwicklungsmethodik für
mechatronische Systeme

Design methodology for
mechatronic systems

VDI 2206

Ausg. deutsch/englisch
Issue German/English

From the mechanical brake to the mechatronic brake in twelve decades



Brake by wire (SIEMENS)

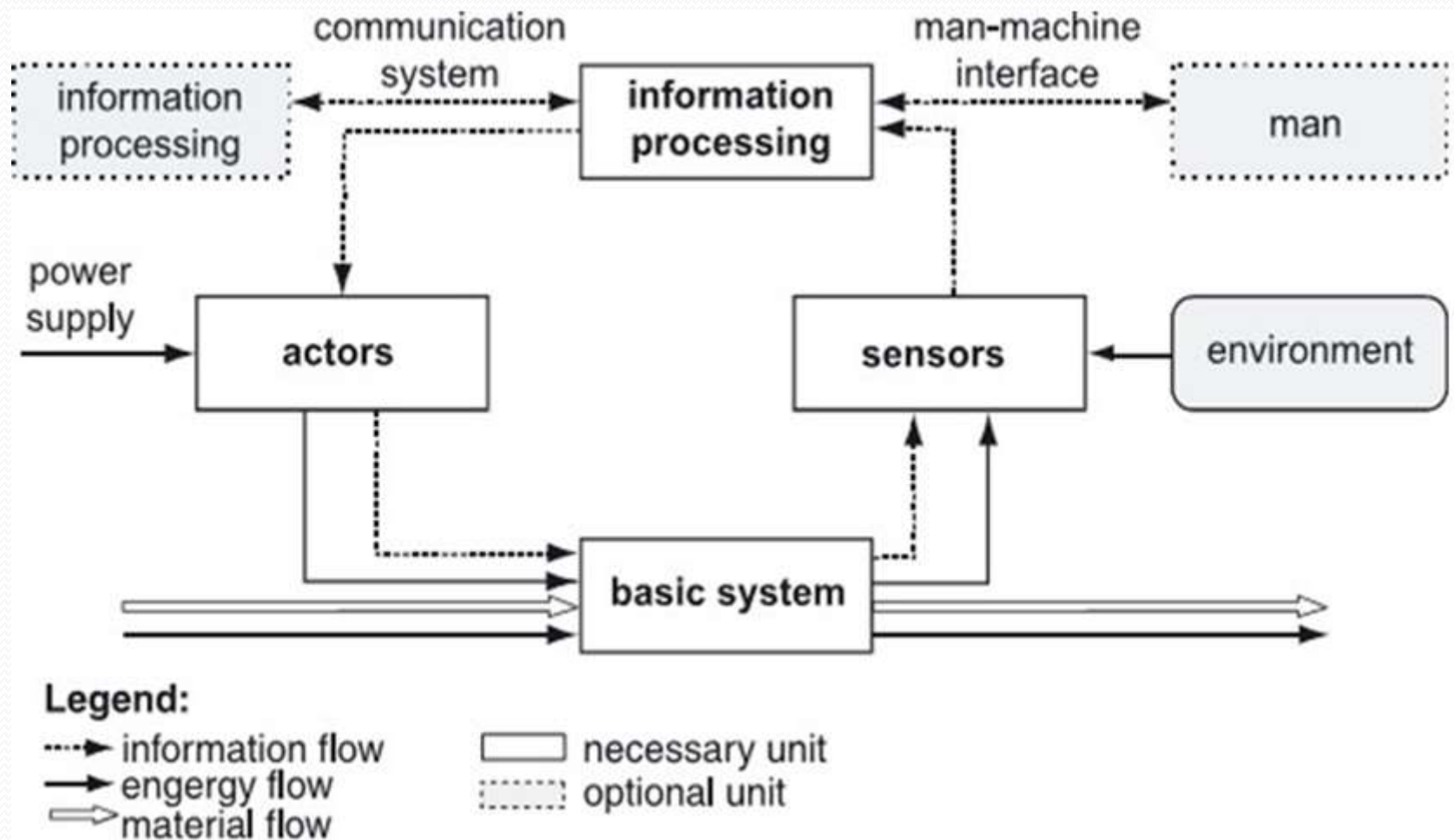
Objective

- ❑ The objective of this guideline is to:
 - provide methodological support for the cross-domain development of mechatronic systems.
- ❑ The main aspects here are intended to be *the procedures, methods and tools for the early phase of development*, concentrating on system design. The result of system design is the assured concept of a mechatronic system.
- ❑ This is understood as meaning the solution established in principle and checked by verification and validation.



Introduction to the Development of Mechatronic Systems

Structure of Mechatronic Systems



Example of structuring of mechatronic systems



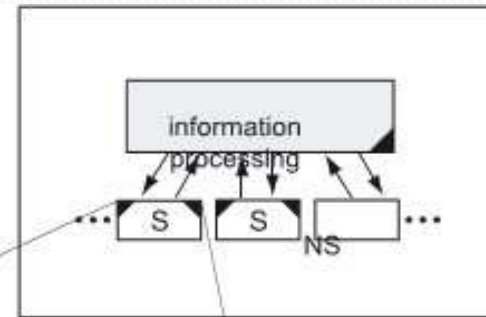
crossing management



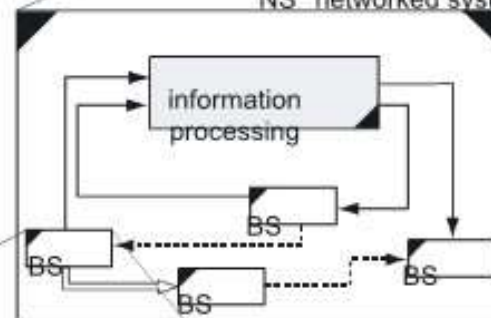
vehicle



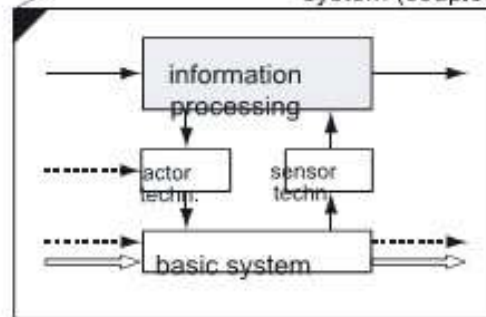
spring strut



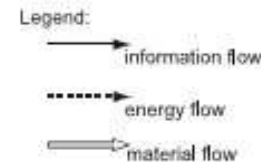
NS networked systems



S: system (coupled basic modules)



BS Basic system





Development Methodology of Mechatronics

Design Procedures

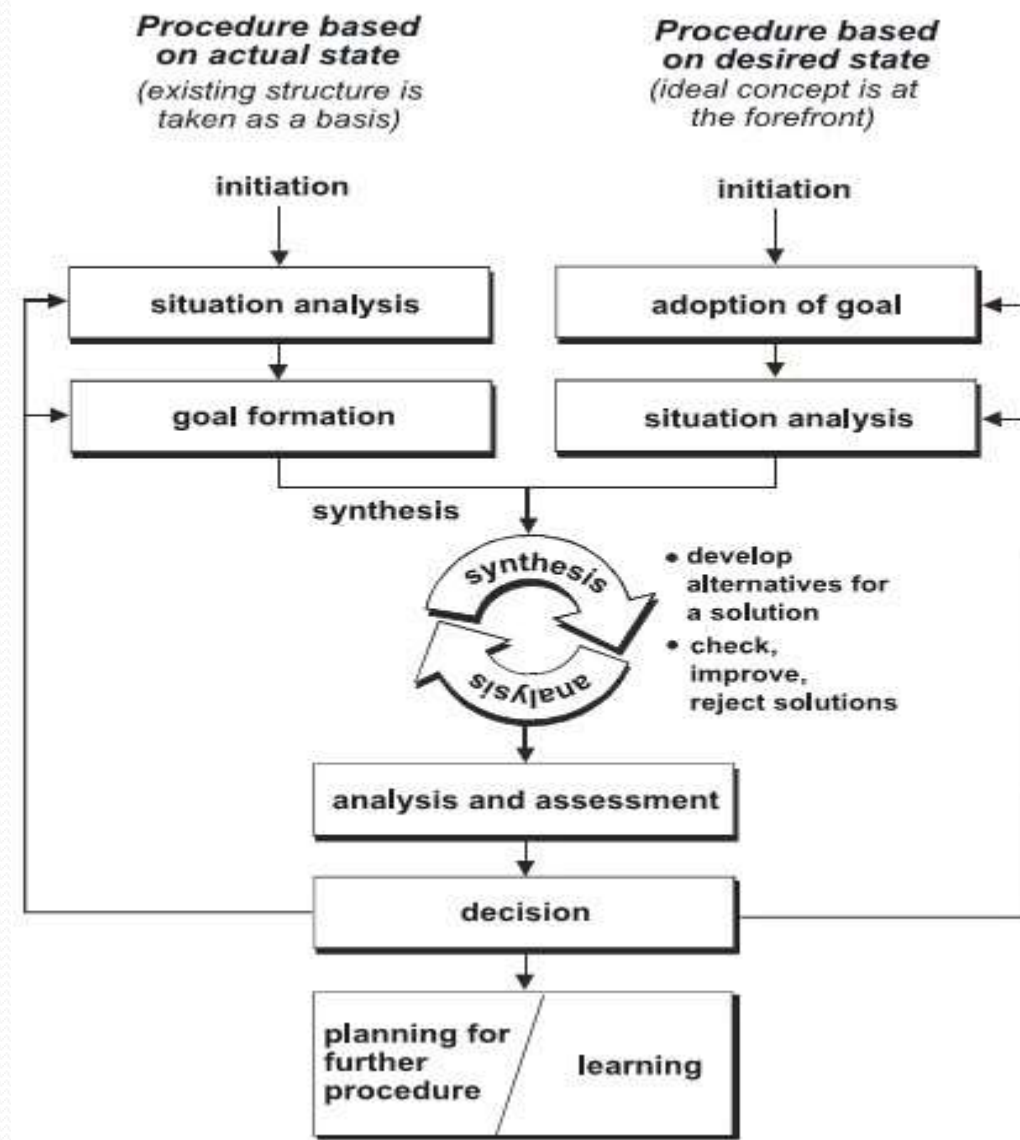
❑ Main procedures:

- Requirements.
- System design.
- Domain specific design.
- Modeling and model analysis.
- System integration.
- Assurance of properties.

Problem-solving cycle as a micro- cycle

- ❑ **Problem-solving cycle as a micro-cycle:** The structuring of the procedure in the development process takes place in this case on the basis of a general problem-solving cycle, such as that known for example from systems engineering. By arranging procedural cycles in series and one within the other, process planning can be flexibly adapted to the peculiarities of any development task.
- ❑ The micro-cycle of the handling organization presented here originates from systems engineering and has also been adopted in modified forms in other disciplines, such as for example business management or software engineering. Its validity in principle for the planning and implementation of effective problem-solving behavior has in this way been confirmed over and over again, including from a psychological aspect. It comprises the following steps:

Problem-solving cycle as a micro- cycle



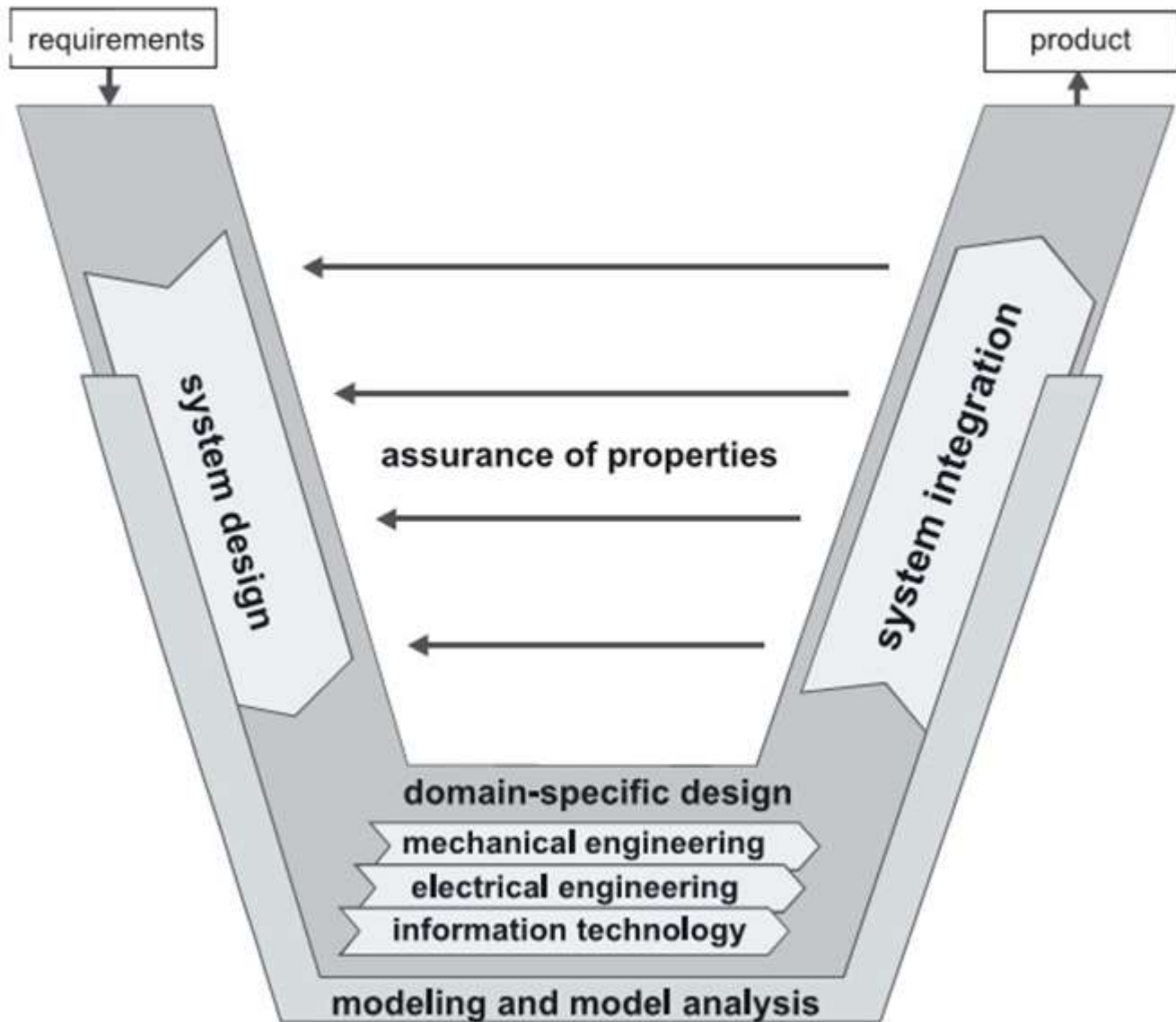
Problem-solving cycle as a micro- cycle

- ❑ **Situation analysis or adoption of a goal:** at the beginning of an elementary handling cycle there is either the situation analysis or the adoption of a goal.
 - ❑ The acting group or individual can adopt an externally prescribed goal, which is followed by a situation analysis (procedure governed by the desired state), or, following the analysis of an initially unclear situation, itself formulate the goal (procedure governed by the actual situation).
- ❑ **Analysis and synthesis:** The search for solutions to the given problem takes place against the background of situation analysis and objective.
 - ❑ This process takes the form in practice of a permanent alternation between synthesis steps and analysis steps which the product developers carry out partly consciously, partly also subconsciously.
 - ❑ The aim of this substep is to work out alternative solution variants.

V Model

- The V model describes the generic procedure for designing Mechatronics systems, which is to be given a more distinct form from case to case

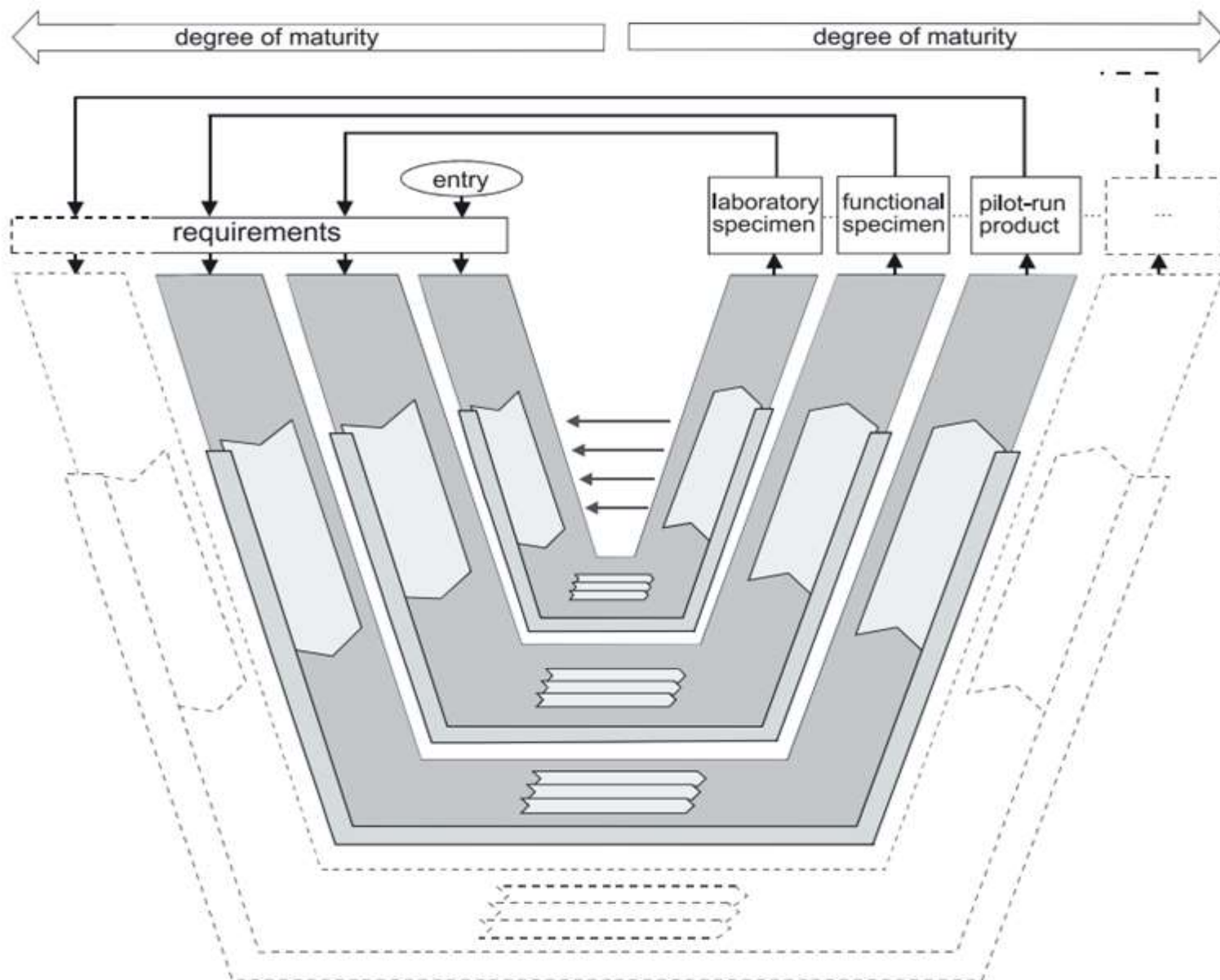
The V model as a macro-cycle: A guide for the basic procedure is offered by the V model adopted from software development and adapted to the requirements of mechatronics; it describes the logical sequence of important substeps in the development of mechatronic systems [Brö95; FKM00].

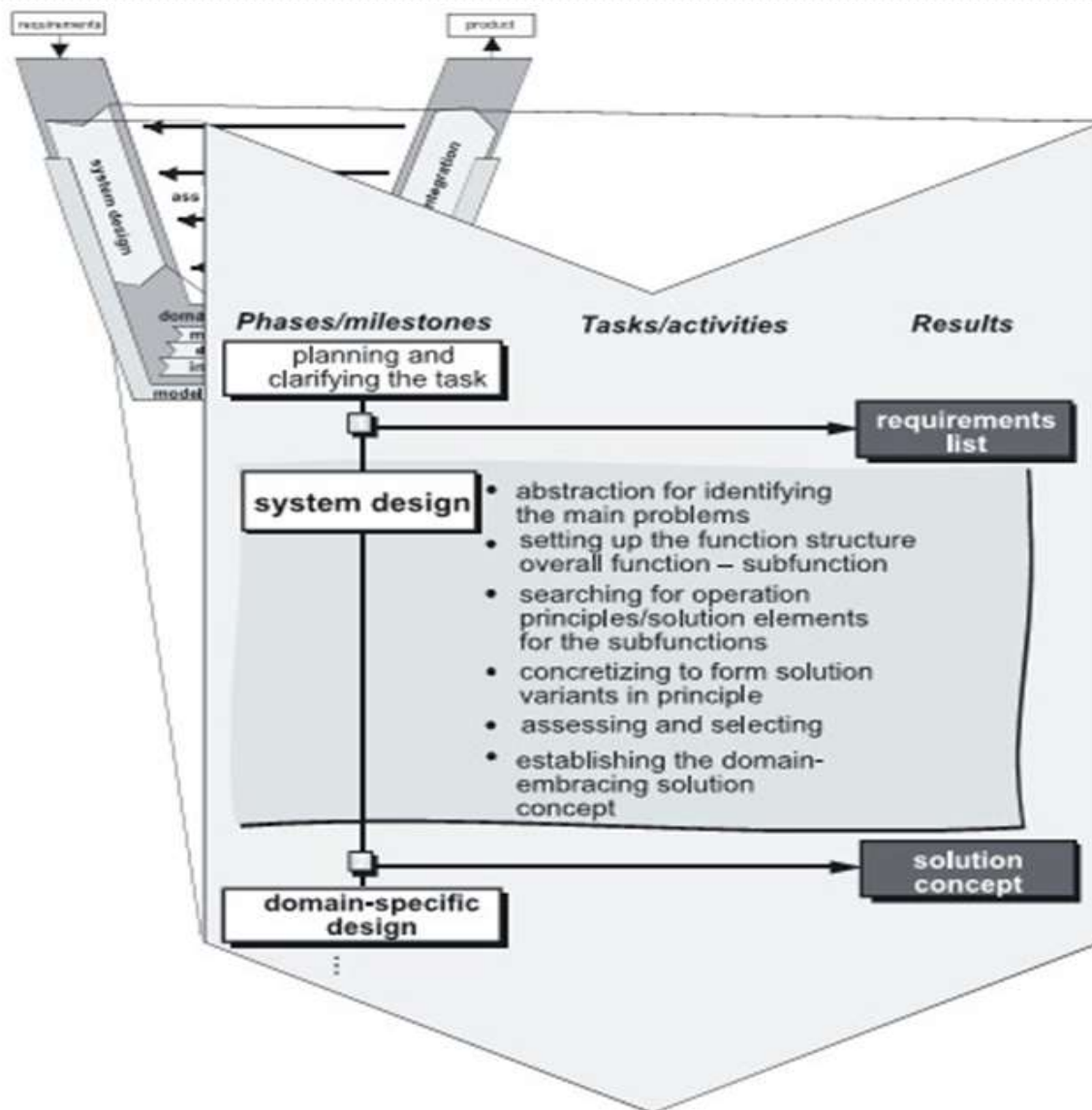




Requirements: The starting point is formed by an actual development order. The defined object was specified more precisely and described in the form of requirements. These requirements at the same time form the measure against which the later product is to be assessed.

System design: The aim is to establish a cross-domain solution concept which describes the main physical and logical operating characteristics of the future product. For this purpose, the overall function of a system is broken down into main subfunctions. These subfunctions are assigned suitable operating principles or solution elements and the performance of the function is tested in the context of the system.



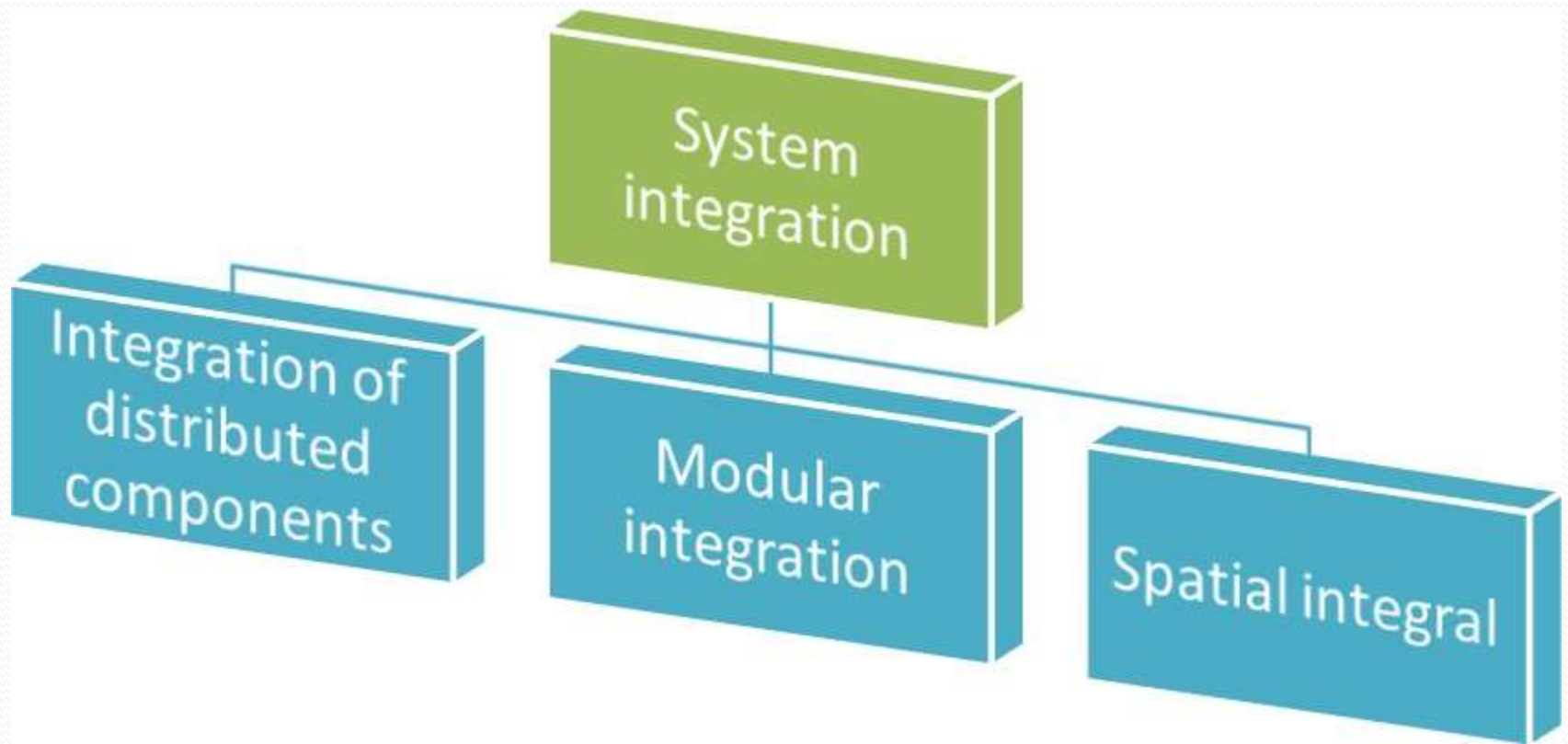


Activities in system design



Domain-specific design: On the basis of this jointly developed solution concept, further concretization usually takes place separately in the domains involved. More detailed interpretations and calculations are necessary to ensure the performance of the function, in particular in the case of critical functions.

System integration: The results from the individual domains are integrated to form an overall system, to allow the interaction to be investigated.



System integration

Integration of distributed components

- Components such as sensors and power actuators are connected to one another via signal and energy flows with the aid of communication systems, that of the energy flows via coupling and plug-in connectors.

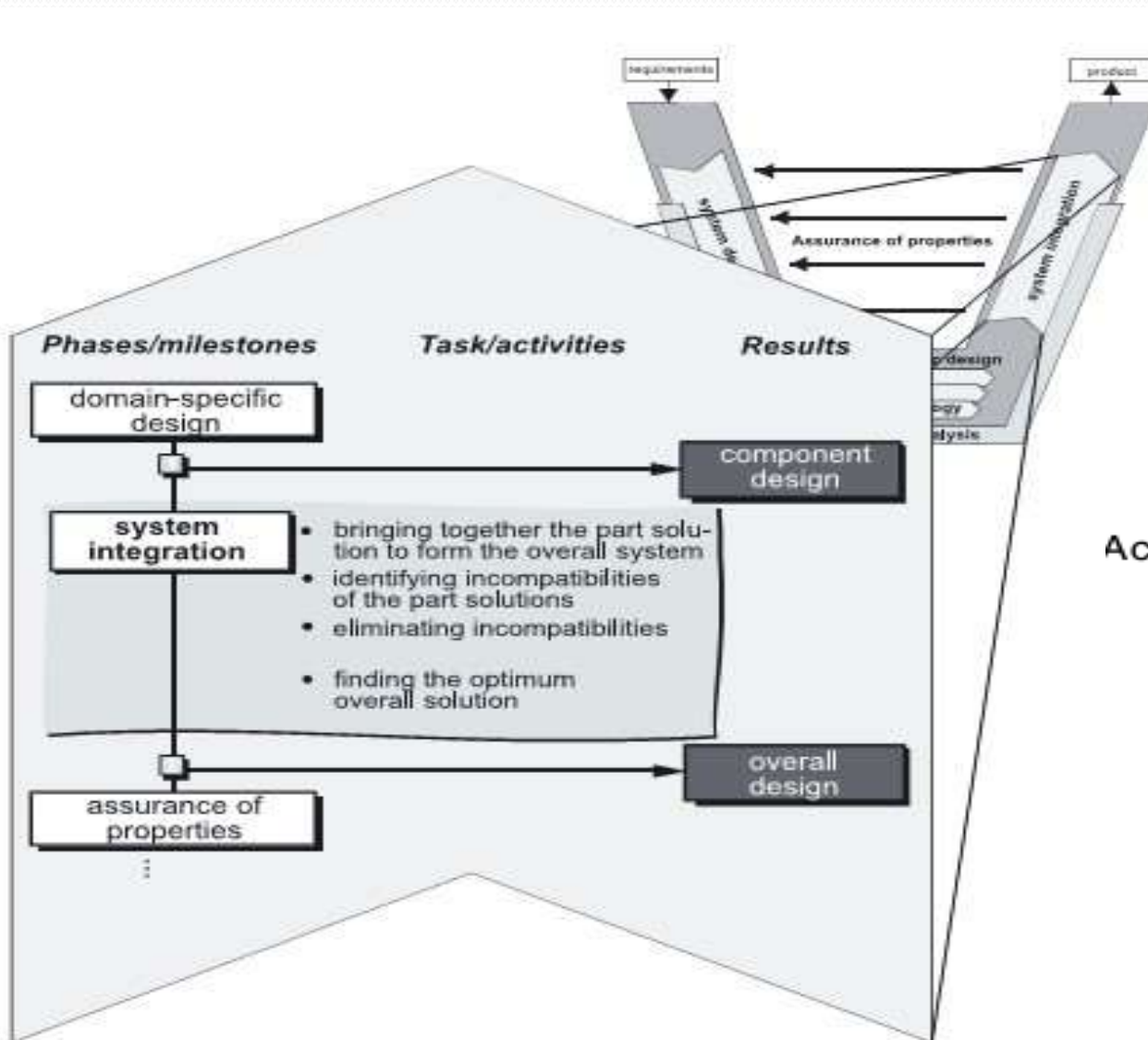
Modular integration

- The overall system is made up of modules of defined functionality and standardized dimensions . The coupling takes place via unified interfaces such as for DIN plug and socket connection , standardized integral.

Spatial integration

- All components are spatially integrated and form a complex functional unit, for example integration of all elements of a drive system (controller, power actuator, motor, transfer element, operating element) into a housing .

System integration





Hardware-in-the-loop (HIL) is the integration of real components and system models in a common simulation environment. The HIL replication (simulation) of dynamic systems by physical and mathematical models must in this case take place in real time and with the physical loads replicated. An example is the simulation of an entire vehicle on a computer with the connection of a real control device and the actor technology for functional control to provide vehicle stability. A decisive advantage of HIL is the function test of the control device under real conditions while at the same time saving on time- and cost-intensive driving maneuvers.

Software-in-the-loop (SIL) is the integration of system models in a common simulation environment with the modeled process (controlled system); both the function to be developed and the process (controlled system) on which the function acts are modeled. The SIL replication (simulation) of dynamic systems by mathematical models does not have to take place in real time. A decisive advantage of SIL is the function test under simulated conditions while at the same time saving on time- and cost-intensive experiments (for example driving maneuvers). On the basis of an SIL environment, either the function, the process or both parts can be physically realized and analyzed with regard to their behavior in a closed loop.



Assurance of properties: The progress made with the design must be continually checked on the basis of the specified solution concept and the requirements. It must be ensured that the actual system properties coincide with the desired system properties.

Assurance of Properties

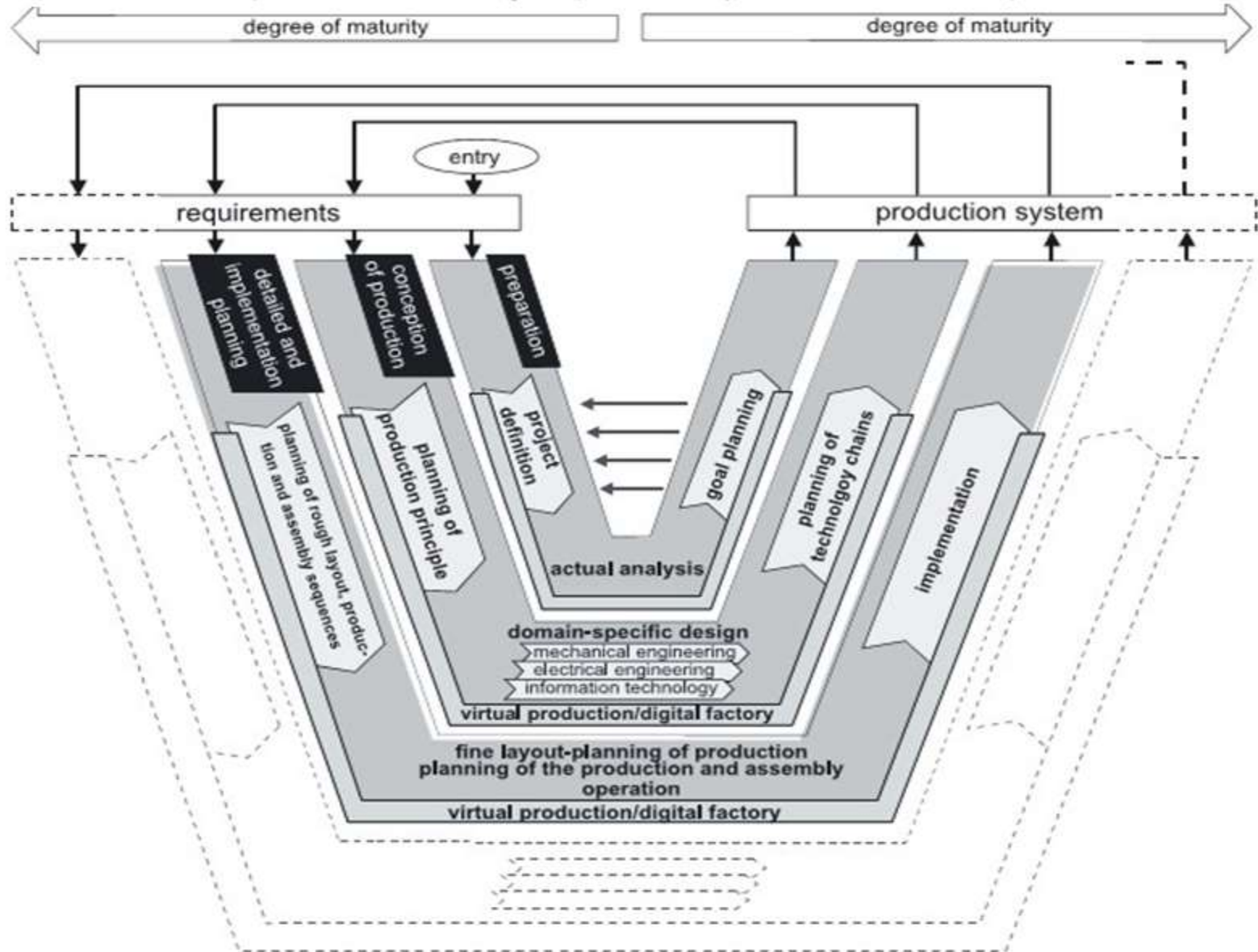
Verification

- Verification means checking whether the way in which something is realized and whether it coincides with the specification.
- Verification is the answer to the question : Is a correct product being developed? For example, does a software program coincide with the deception of algorithms.

Validation

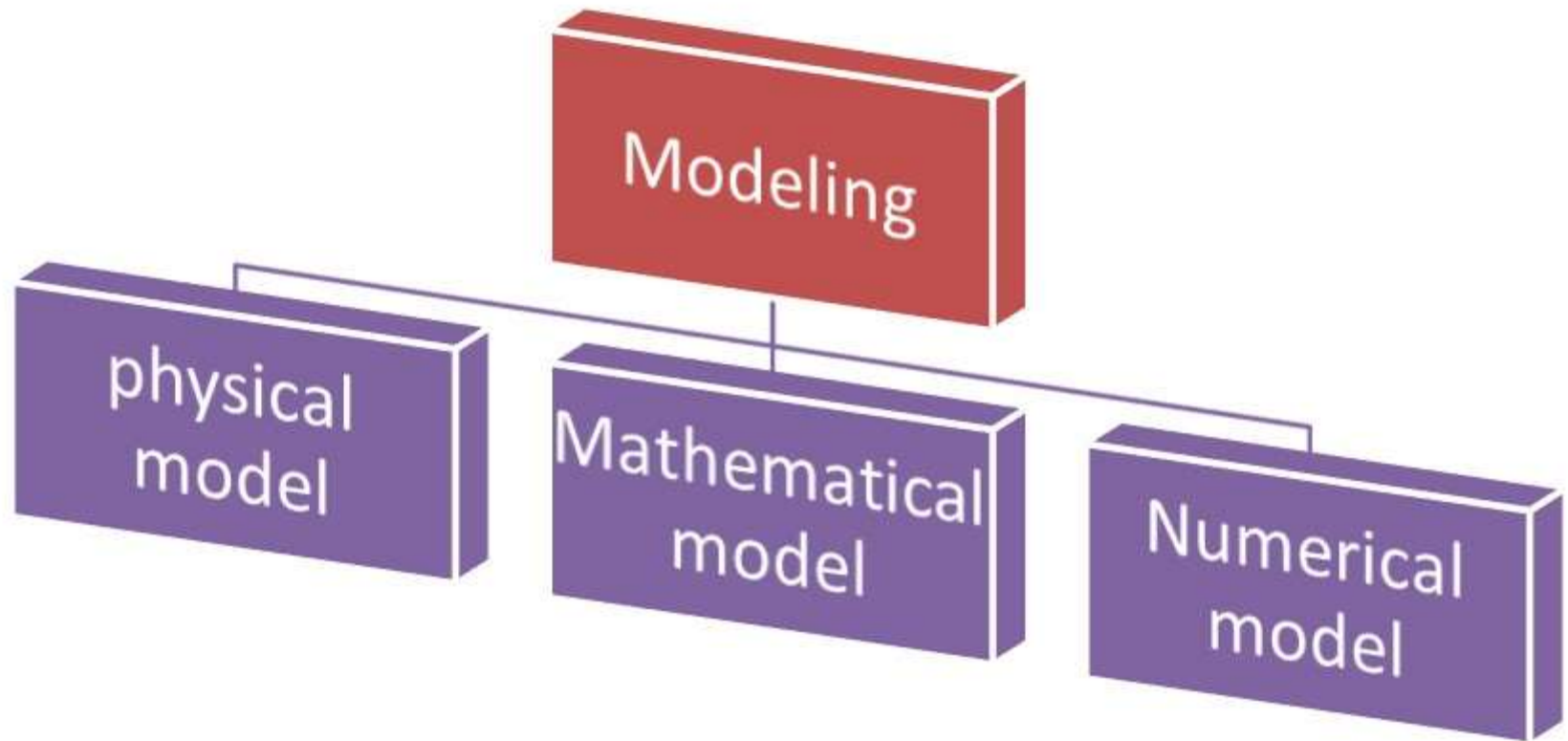
- Validation means testing whether the product is suitable for its intended purpose or achieves the desired value.
- Validation is the answer to the question : Is a right product being developed?

iterative procedure in the design of production systems for mechatronic products





Modeling and model analysis: the phases described are flanked by the forming and investigating of the system properties with the aid of models and computer-aided tools for simulation.



Integration of distributed components

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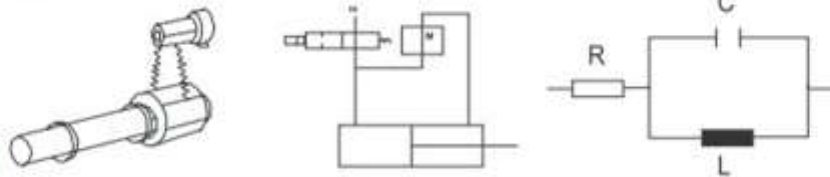
Modular integration

- The overall system is made up of modules of defined functionality and standardized dimensions . The coupling takes place via unified interfaces such as for DIN plug and socket connection , standardized integral.

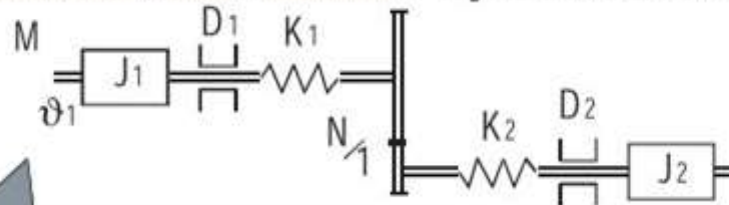
Spatial integration

- All components are spatially integrated and form a complex functional unit, for example integration of all elements of a drive system (controller, power actuator, motor, transfer element, operating element) into a housing .

Topologisches Modell/Topological model



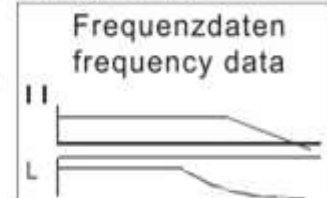
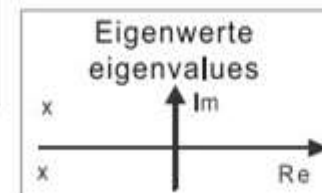
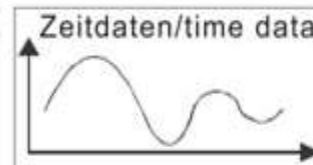
Physikalisches Modell/Physical model



Mathematisches Modell/Mathematical model

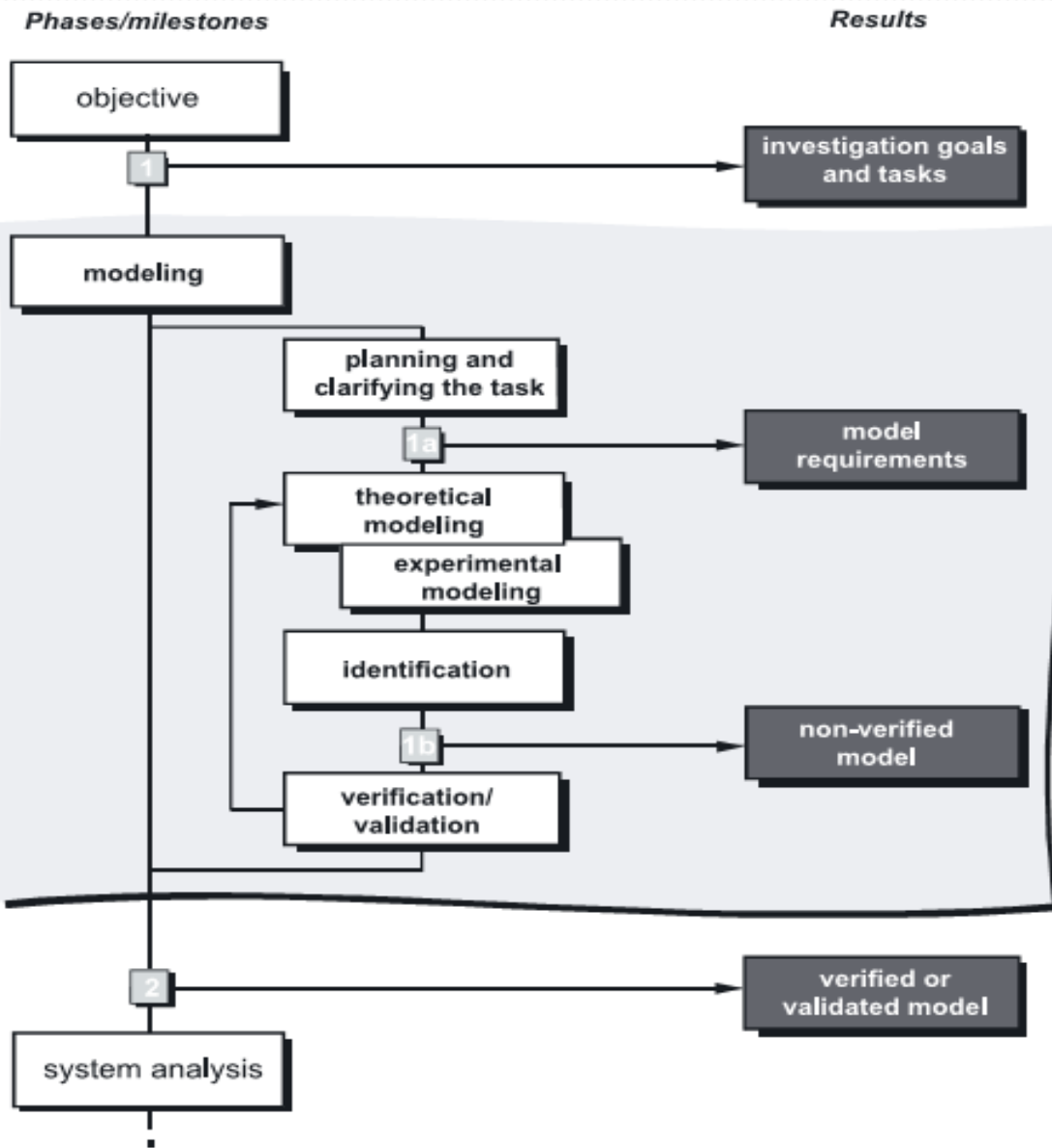
$$(J_1 + N_2 J_2) \ddot{\vartheta}_1 + (D_1 + N_2 D_2) \dot{\vartheta}_1 + (K_1 + N_2 K_2) \vartheta_1 = M$$

Numerisches Modell/Numerical model



Legende/Legend:

- ϑ Drehwinkel/rotating angle
- C elektr. Kapazität/el. capacitance
- D Dämpfungsmaß/degree of damping
- J Massenträgheitsmoment/mass moment of inertia
- K Drehsteifigkeit/rotational rigidity
- L Induktivität/inductance
- M antreibendes Drehmoment/driving torque
- N Übersetzung/transmission
- R elektr. Widerstand/electrical resistance



Procedure for modeling

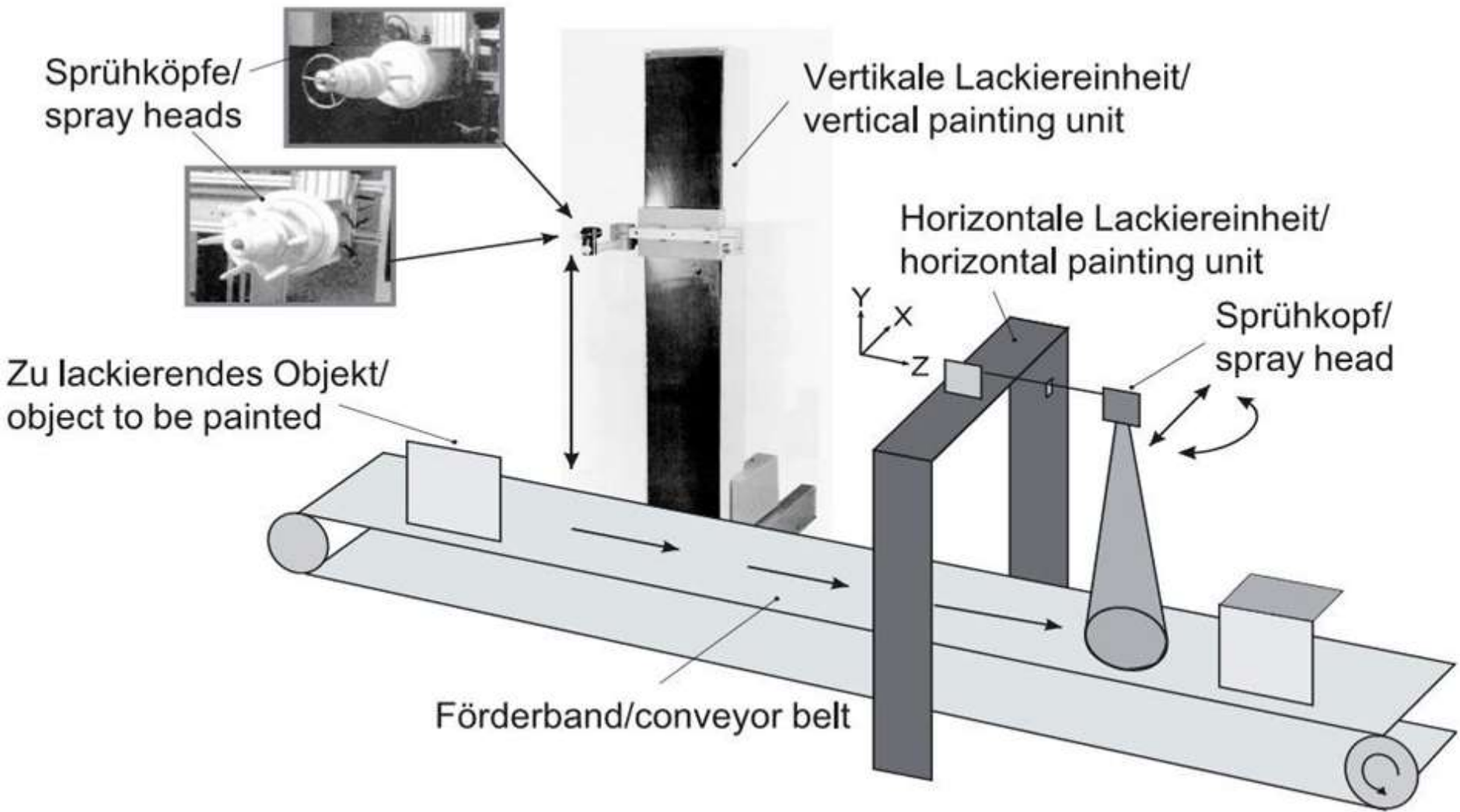


Product: The result of a continuous macro-cycle is the product. In this case, a product is understood as meaning not exclusively the finished, actually existing product but the increasing concretization of the future product (product maturity). Degrees of maturity are, for example, the laboratory specimen, the functional specimen, the pilot-run product, etc.

A complex mechatronic product is generally not produced within one macro-cycle. Rather, a number of cycles are required (Figure 3-3).

4.2 Design of the drive unit of a simple painting system

For painting mass-produced articles (kitchen appliances, audio and video equipment, aluminum wheel rims), painting systems in the form of continuous lines are often used. On these, the objects to be painted pass continuously through the system on a conveyor belt. The paint is applied by a number of spraying units, the oscillating movement of which runs either vertically (for the side surfaces of the object) or horizontally (for the upper side of the object) (Figure 4-11).



Basic representation of the continuous line

System design

- search for variants for converting a rotational movement into a translatory movement
- rough dimensioning to the mechanical and electrical drive elements (power, mass, rigidities, inductances, torque constant, etc.)
- assessment and selection of variants

modeling : system analysis

- Modeling of the overall system (Matrix X/SystemBuild model)
- design of the controller structure or the control
- checking the realizable requirements in the simulation model

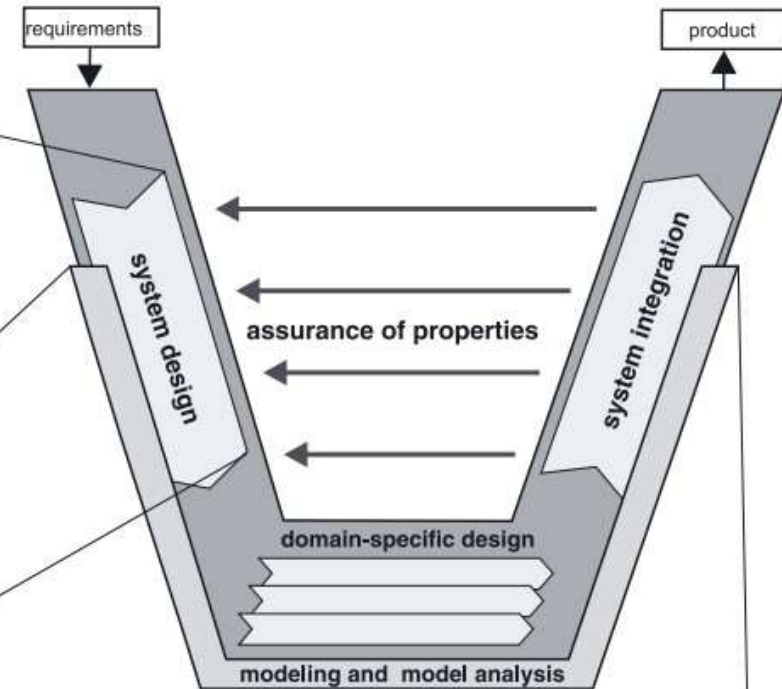


Fig. 4-10. Procedure for designing the drive unit of a simple painting system

4.2.1 Selected requirements

Desired speed in x direction:

1 m/s

Distance at constant desired speed in x direction over the belt:

700 mm (ideally: 900 mm)

Maximum oscillating amplitudes in x direction in the range of constant desired speed:

± 3 mm (ideally: ± 2 mm)

Electric drive

The geometrical requirements are summarized in Figure 4-12.

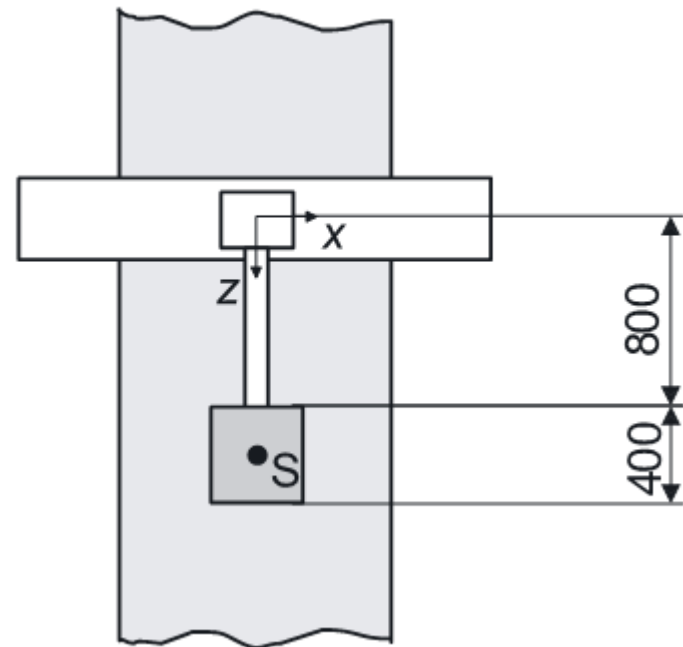
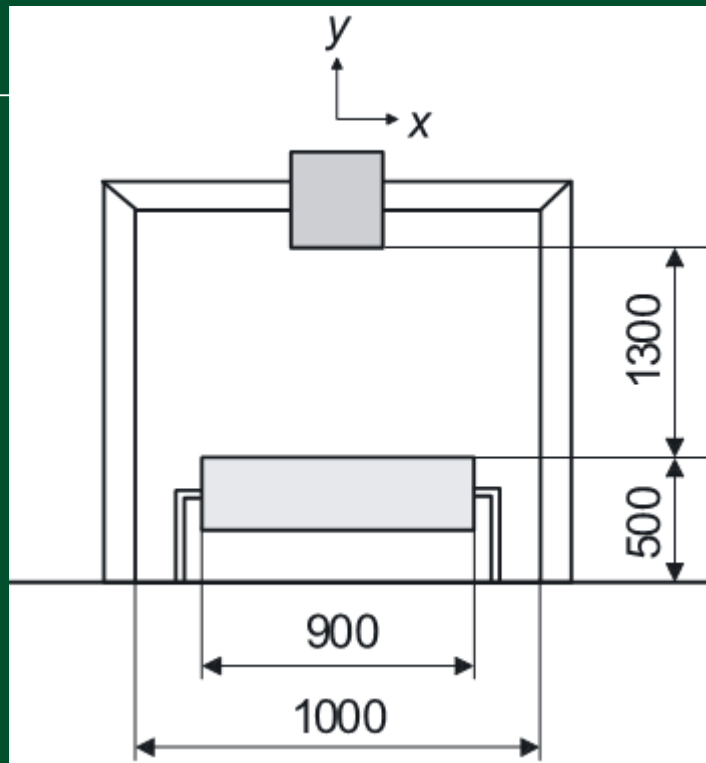


Fig. 4-12. Geometrical requirements

Direkte Erzeugung der Translation/ Translation produced directly

Transformation von Rotation in Translation/ Transformation from rotation into translation

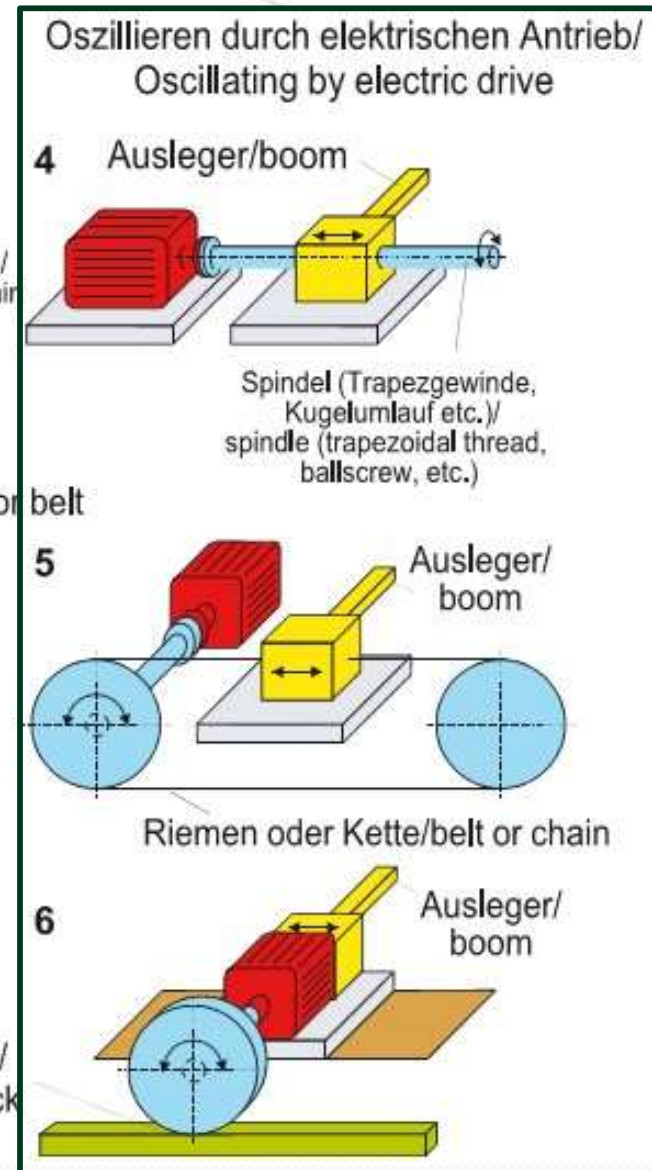
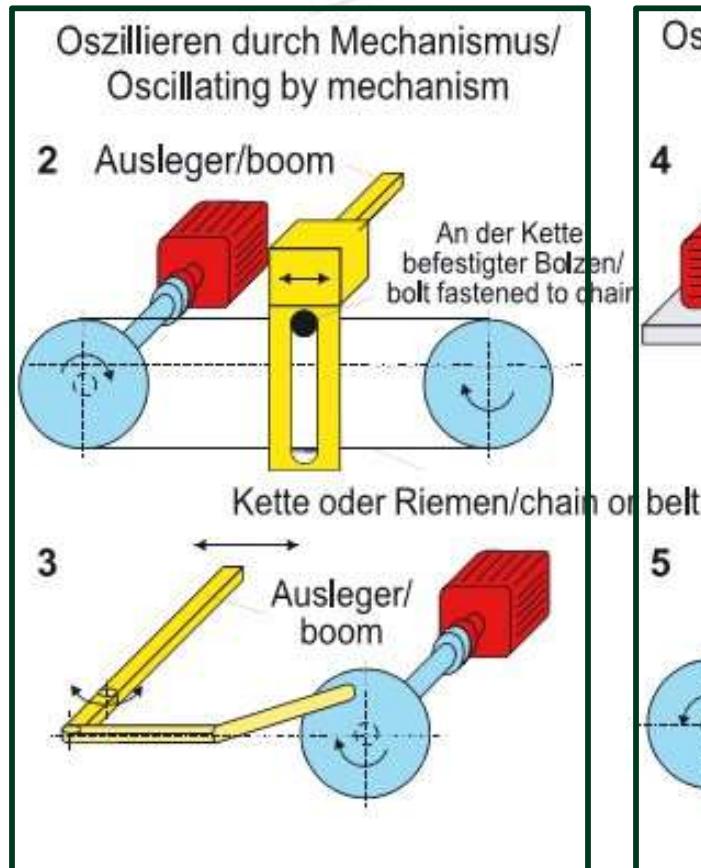
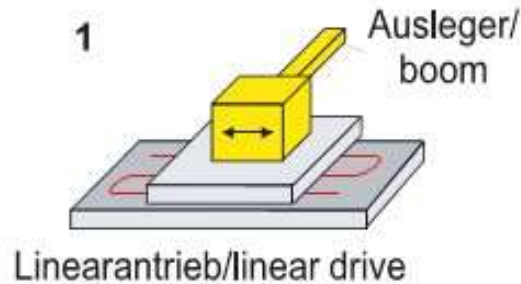
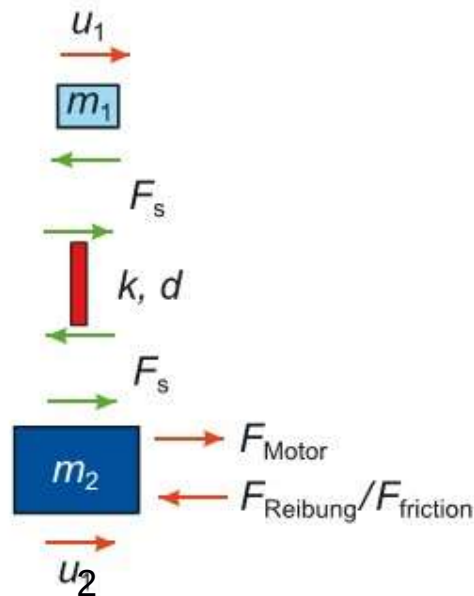
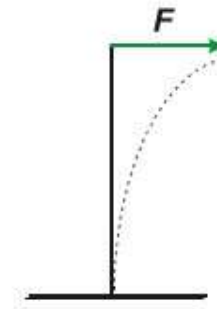
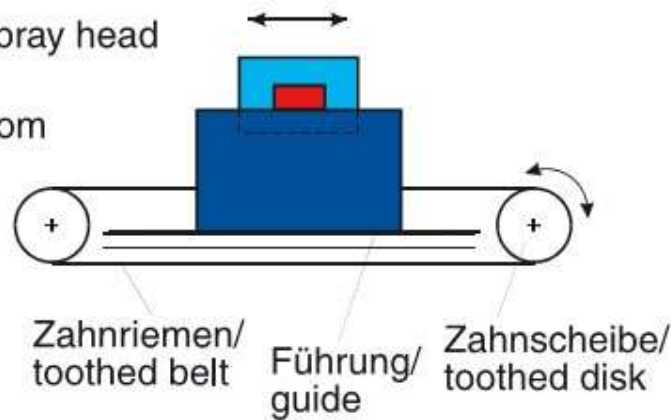
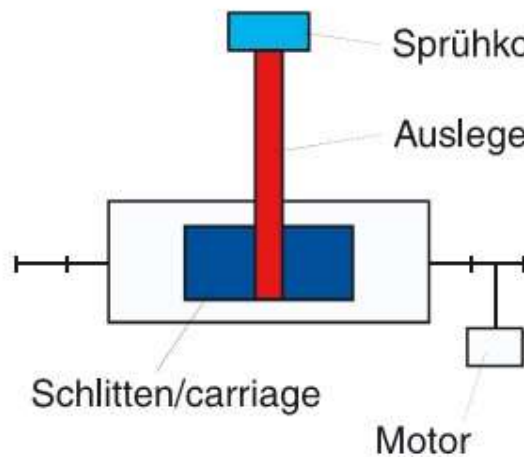


Fig. 4-13. Solution variants of the system design



A more precise assessment of the variants cannot be made within the scope of this guideline. However, it should be noted that variants of type 2 represent a solution that is customary in practice, but have great problems with mechanical loads and wear in the connecting link guide and also lead to inflexible kinematics in the returning area. Therefore, variant 5 with a brushless DC motor in combination with a toothed belt drive is to be considered by way of example.



$$-m_1 \cdot \ddot{u}_1 - F_s = 0$$

$$F_s = F_k + F_d = k \cdot (u_1 - u_2) + d \cdot (\dot{u}_1 - \dot{u}_2)$$

$$u = \frac{F \cdot l^3}{3EI} \rightarrow k = \frac{3l}{1}$$

$$-m_2 \cdot \ddot{u}_2 + F_s + F_{\text{Motor}} - F_{\text{Reibung}} = 0$$

$$F_{\text{Motor}} = 2 \cdot M_{\text{el}} / d_{\text{Zahnscheibe}}$$

Figure 4-14. Formulating the differential equations for the simple mechanical substitute model

The DC motor used is described by its winding inductance and winding resistance and also two constant factors k_i and k_E with the equations

$$i = \frac{1}{L} \cdot (U_{\text{motor}} - R \cdot i - k_E \cdot \dot{\phi})$$

$$M_{\text{el}} = k_i \cdot i$$

The term $k_E \cdot \dot{\phi}$ describes the mutual induction and couples the mechanical and electrical equation systems.

4.2.4 Control structure

The system is designed as a cascade control with three PID controllers. In the inner cascade, the current control ensures that the inductance of the motor winding which limits the rate of current rise is compensated by an increased motor voltage. The dynamic behavior of the motor is consequently idealized. The middle cascade comprises the speed control of the motor, while the outer cascade controls the position. The desired displacement is also additionally fed forward here as a precontrol. Furthermore, limitations both for the maximum motor current (43 A) and the maximum motor voltage (300 V) are integrated in the controller cascade. Figure 4-15 shows the overall system.

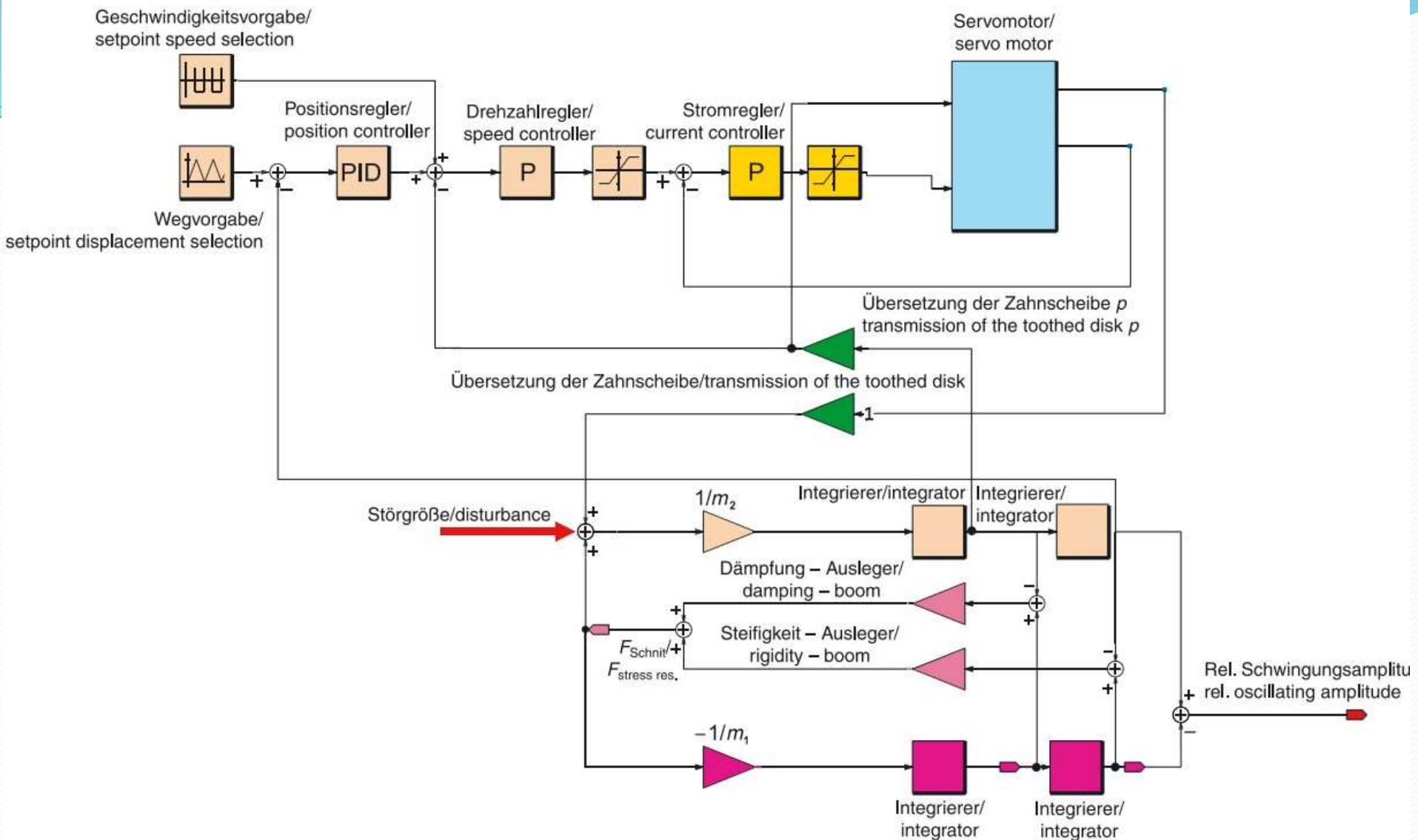


Fig. 4-15. Block diagram of the overall system

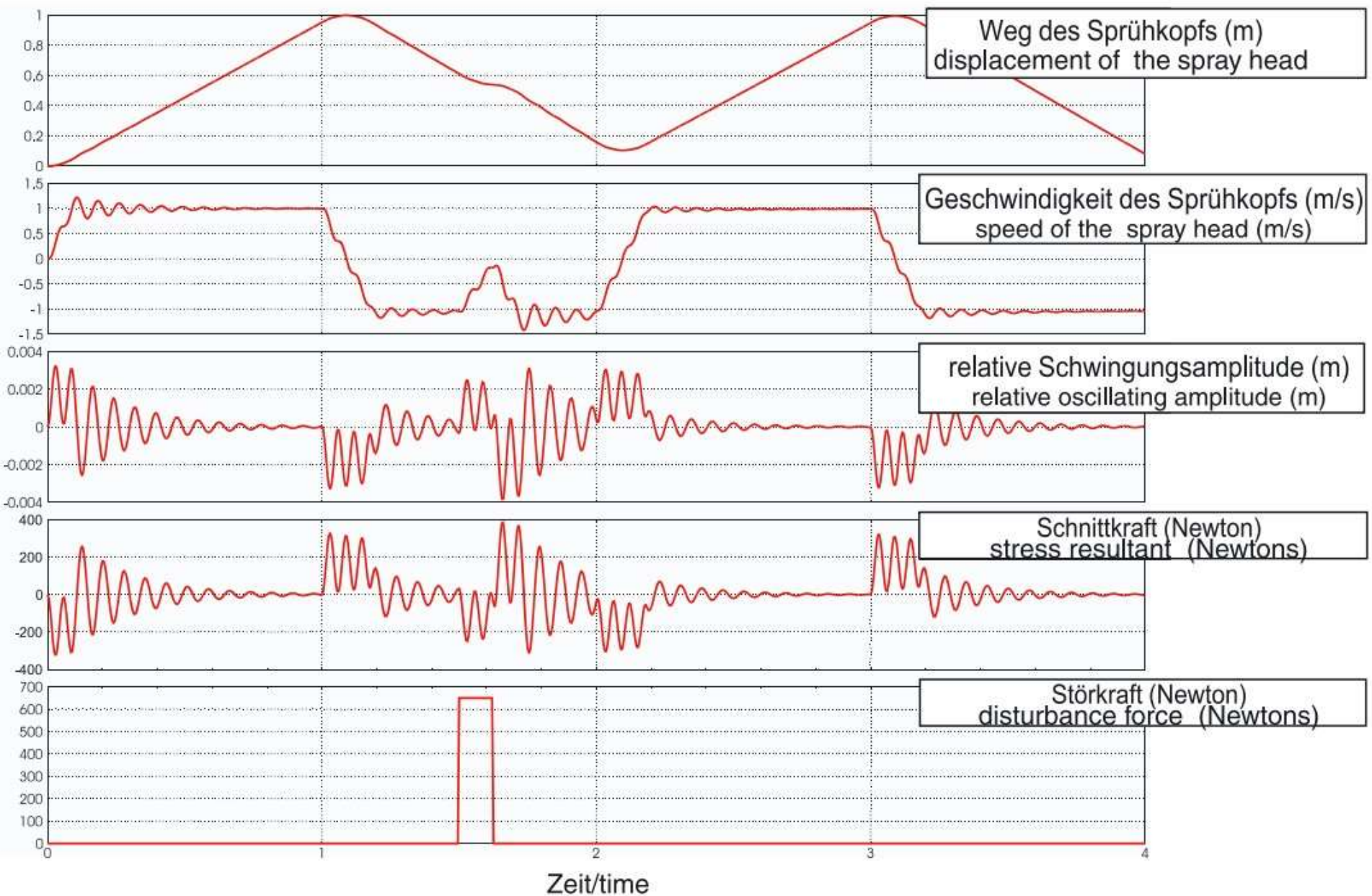


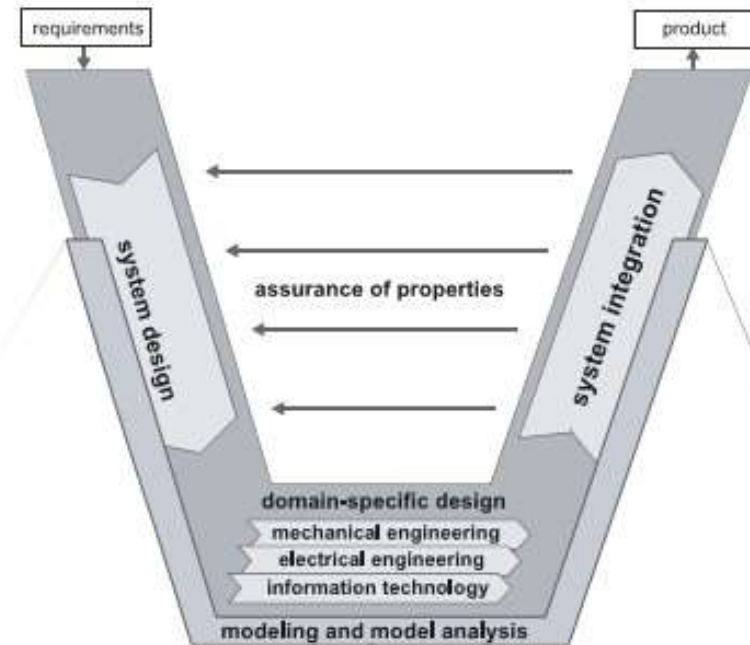
Fig. 4-16. Simulation results of the block diagram from Figure 4-15

Design of an active spring/tilting module

Problem Definition:

- ❑ This example shows the design of an **active chassis in railroad technology**.
- ❑ There follows a description of the **hierarchical structure and the control**.
- ❑ Improving **riding comfort and safety** are important requirements for **modern rail transportation technology**. **Conventional rail vehicles** are equipped with a **passive spring-damper combination** and also have **poor riding comfort** in comparison with today's vehicle technology.
- ❑ Faults in the level of the track bed lead to **car body oscillations**, which **impair the riding comfort** of the passenger and put safety at risk.
- ❑ These chassis properties can be **improved** with **the aid of active suspension technology**.
- ❑ High-speed travel over track bends is also to be achieved by the use of an **active tilting device**.

The procedure in the phases of **modeling to system analysis** when designing an active spring/tilting module

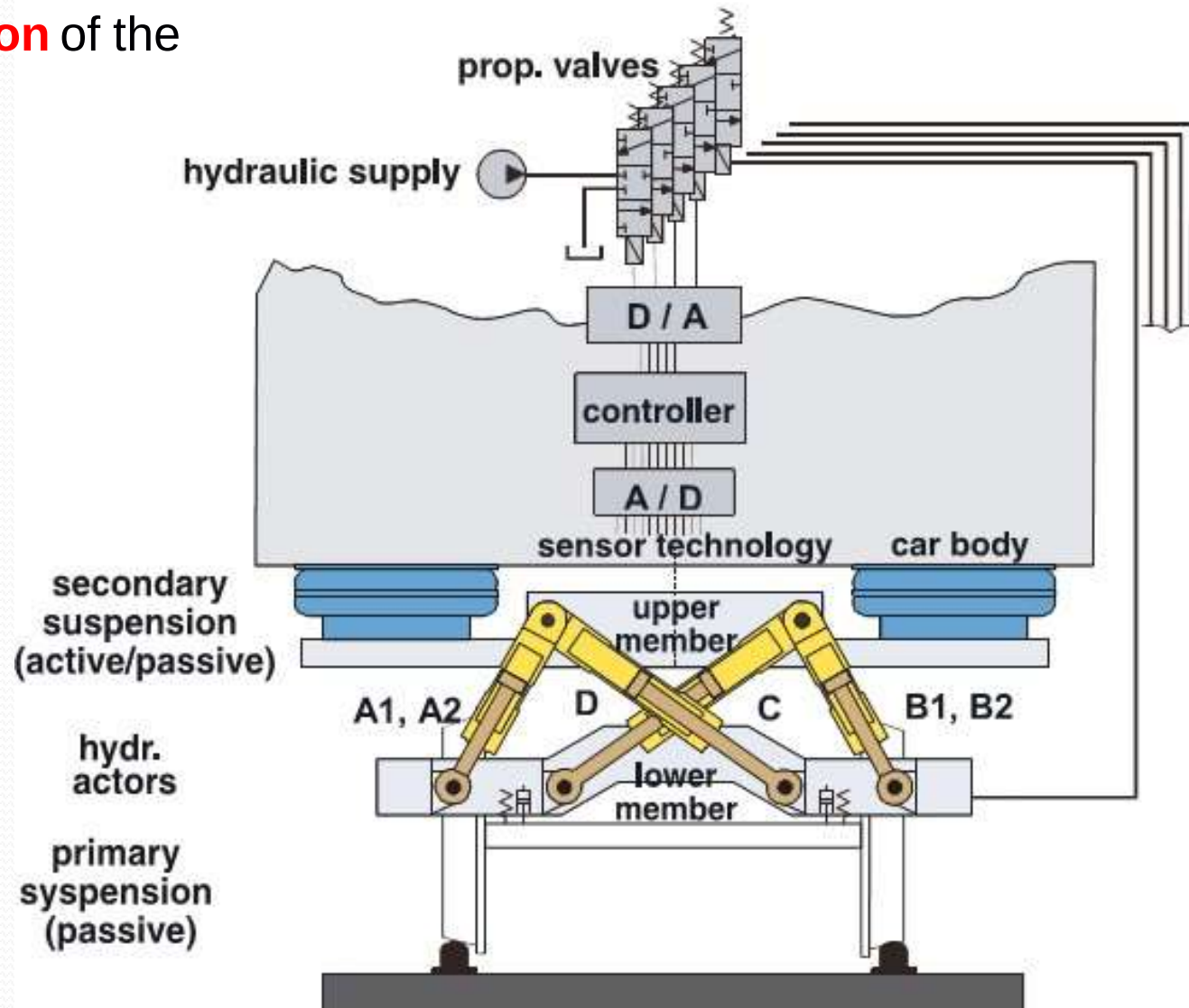


**modeling
:
system analysis**

- function-oriented design of the system structure and hierarchy
- design of the controller structure and modeling of the overall system (multibody system model)
- design of the hierarchical-modular controller for active influencing of the construction (cascade controller)
- trying out the overall controlled system on the HIL (hardware-in-the-loop) test stand

Design of an active spring/tilting module

The **basic construction** of the spring/tilting system



Design of an active spring/tilting module

Basic construction

- ❑ By contrast with conventional chassis, it is intended to *dispense with all the passive dampers of the secondary suspension* of conventional rail chassis.
- ❑ The car body is connected to the chassis only by means of the **pneumatic springs**. While the *pneumatic spring isolates the vibrations in the upper frequency range*, the desired damping in the lower frequency range is realized by adjusting *the base point of the pneumatic spring above the upper member*.
- ❑ The disturbances introduced as a result of faults in the level of the track bed can hardly be transferred any longer to the car body and very good riding comfort is achieved.
- ❑ The information required for controlling the base point adjustment is provided by suitable sensors and processed in a hierarchically constructed multivariable control.
- ❑ The active tilting device, which permits tilting of the car body into the inside of the curve, can be realized with the same adjusting system.

Design of an active spring/tilting module

Basic construction

- ❑ The core structure of the adjusting system comprises the mechanical components of the upper and lower members or the pneumatic spring, the hydraulic actors and the sensors.
- ❑ While four actors *A1*, *A2*, *B1*, *B2* are responsible for lifting and tilting, the other two actors *C* and *D* primarily take care of the lateral movement. In this case, the local pitching of the chassis plane is prevented by the actors *A1*, *A2* and *B1*, *B2* and the longitudinal and yawing movements are blocked by lemniscate levers.
- ❑ As a result, each spring/tilting module actively ensures three directions of movement of vertical, lateral and tilting movements.
- ❑ The vehicle (comprising two modules) permits all controlled rotational and translatory movements in the lateral and vertical directions. The translatory movement in the longitudinal direction is realized by means of the linear drive.

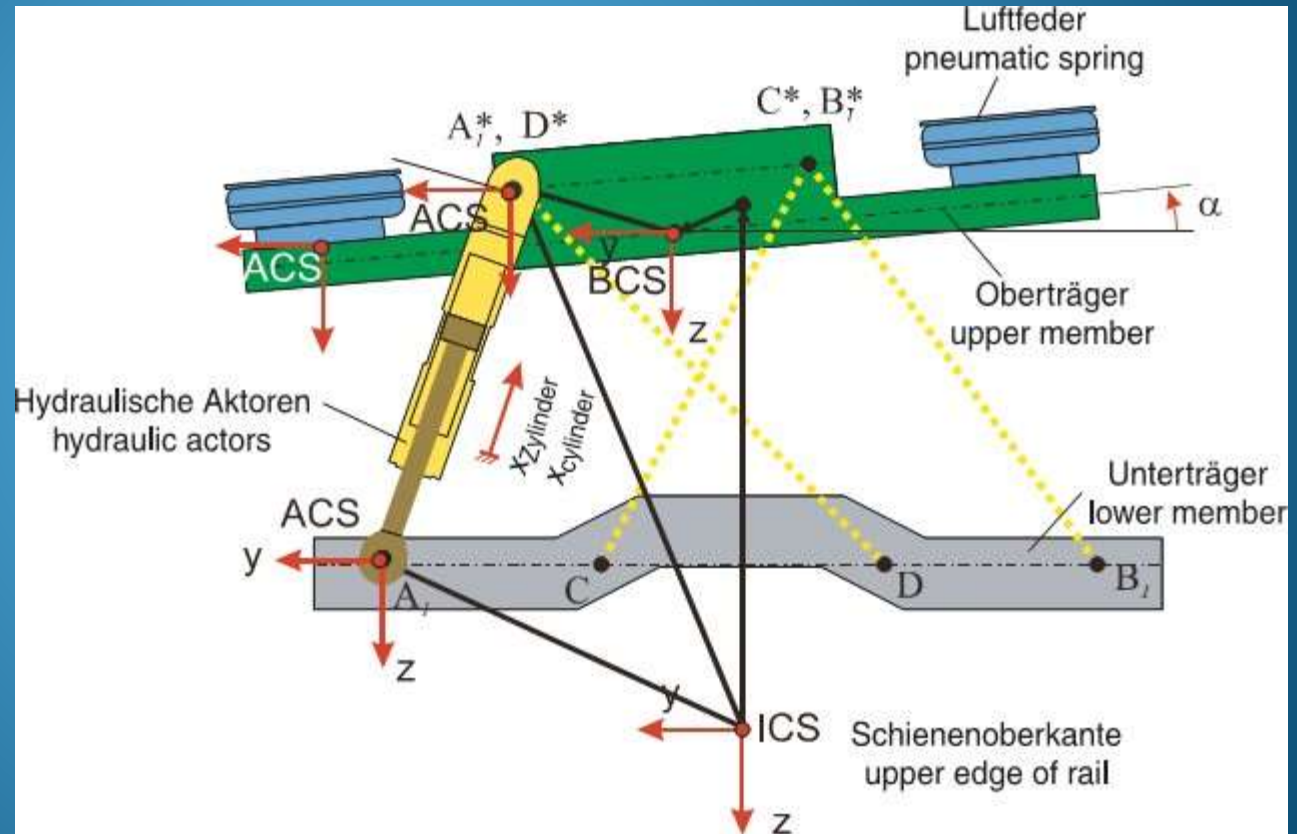
Design of an active spring/tilting module

Modeling

- ❑ In order to investigate the behavior of a dynamic system and subsequently design a multivariable control, first of all the physical substitute model and the mathematical substitute model are formed.
- ❑ The model is intended to represent the **kinematic**, **static** and **dynamic** behavior of the system to be investigated.
 - ❖ Kinematic functions
 - ❖ The kinematic behavior of the system is determined by the degrees of freedom and the geometry of the spring-tilting module.
 - ❖ Dynamic functions

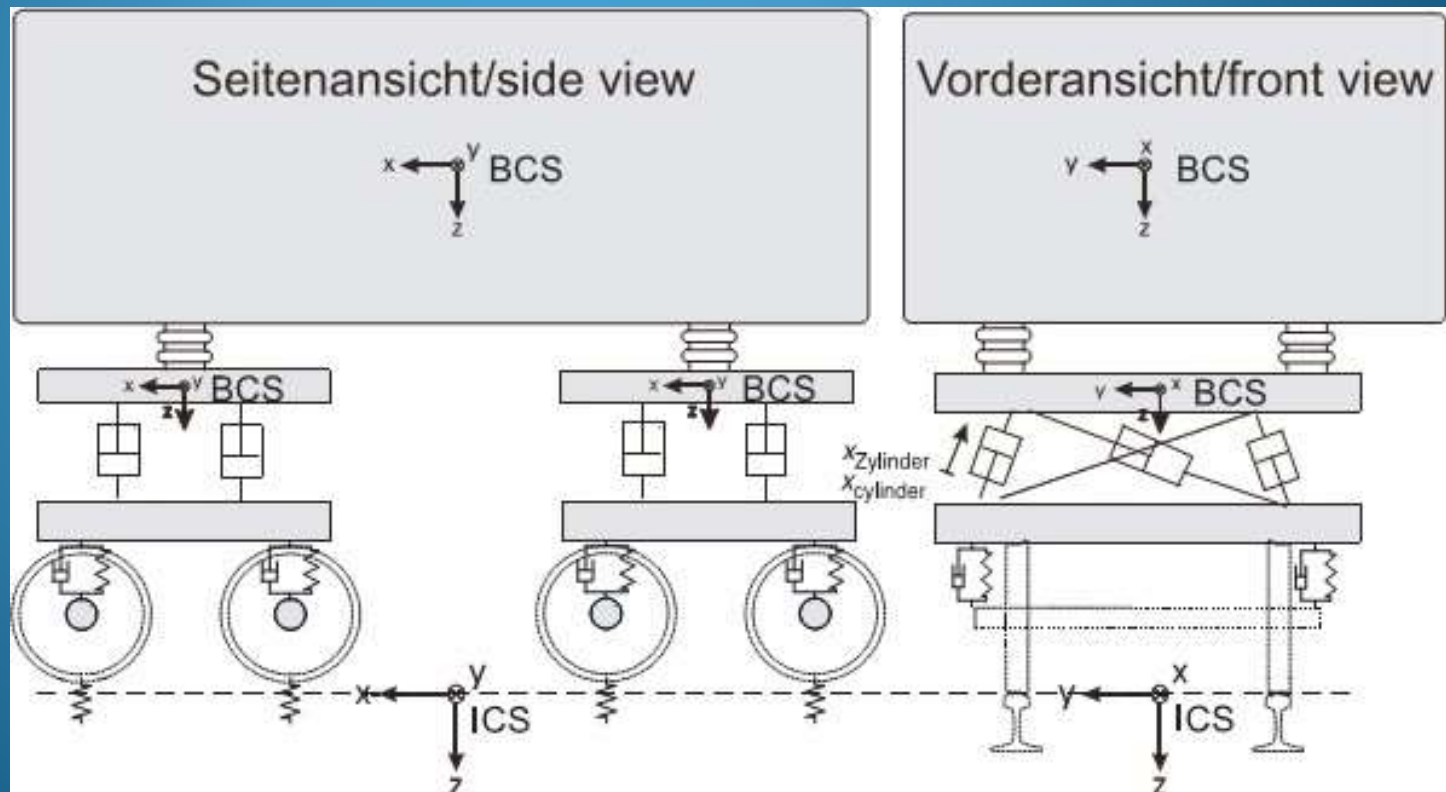
Design of an active spring/tilting module

Coordinate systems of the kinematic function



Design of an active spring/tilting module

Spatial model for investigating the spring/tilting technique



Design of an active spring/tilting module

❑ Mechanical supporting structures

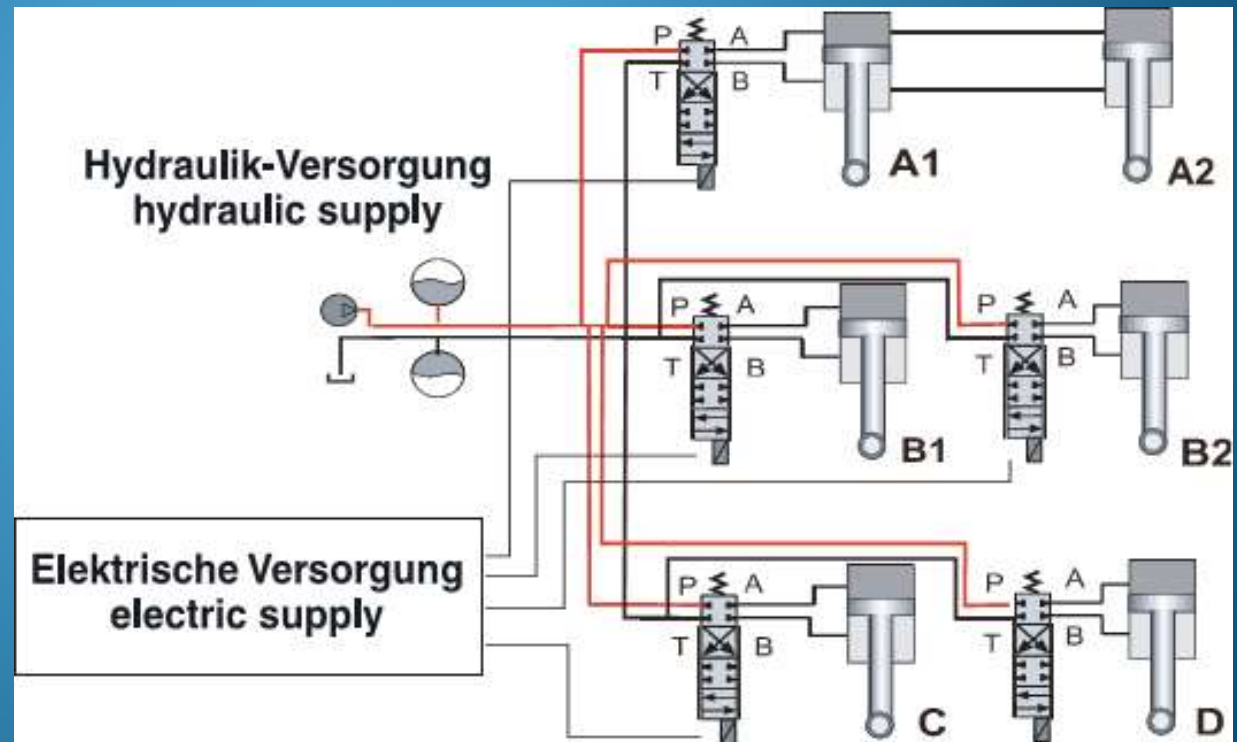
- ❖ The car body and upper member of the spring/tilting module are *modeled as rigid bodies* with **six degrees of freedom** in each case. *The elastic properties of the car body are not taken into consideration* in this investigation, since the natural frequencies of the first three bending and torsion modes of the car body lie in another frequency range.

❑ Dynamics of the adjusting system

- ❖ Hydraulic actor systems are used here. The adjusting system principally comprises **six differential hydraulic cylinders** with **five servo valves**. The cylinder chambers of the cylinders A1 and A2 are connected in parallel and activated by a valve, while the cylinders B1, B2, C and D are respectively activated separately by a valve.

Design of an active spring/tilting module

Schematic setup of the hydraulic actor systems



Design of an active spring/tilting module

- ❑ **Dynamics of the sensor technology and the digital signal processing**
 - ❖ When designing the controller, the dynamic behavior of the sensors used and the dead times occurring with the digital realization of the controller must be taken into consideration in the overall dynamics.
 - ❖ **Inductive displacement** transducers are selected here as position sensors for sensing the cylinder displacement.
 - ❖ The measurement of the position of each valve slide takes place by means of a **position sensor integrated** in the servo valve.
 - ❖ **Linear potentiometers** are used for the measurements of the spring excursions.
 - ❖ The sensors reduce the bandwidth of the system. This effect is modeled by a low-pass filter element.

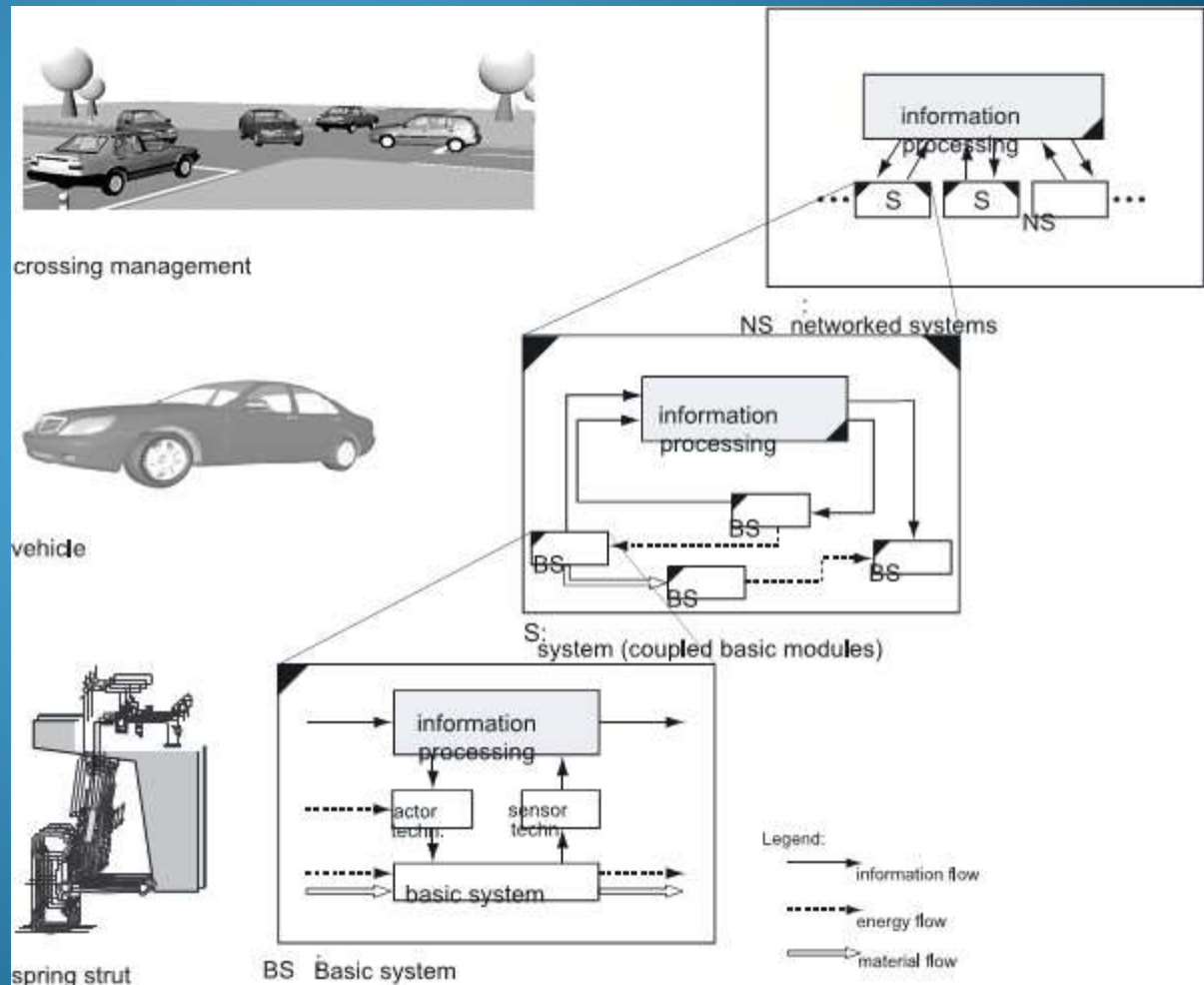
Design of an active spring/tilting module

□ Dynamics of the complete module

- ❖ After the aforementioned investigations, *the dynamics of the complete module are modeled with a development environment*, with the models of all the sub-systems being brought together topologically in the computer with suitable interfaces.
- ❖ The supporting structure is extended by adding the actors, sensors and digital effects, **taking the kinematic interrelationships into consideration.**
- ❖ Once the dynamic behavior of the system has been investigated, control structures must be designed, in order that the mechatronic functions of the system achieve the desired system behavior.

Design of an active spring/tilting module

Example of structuring of mechatronic systems

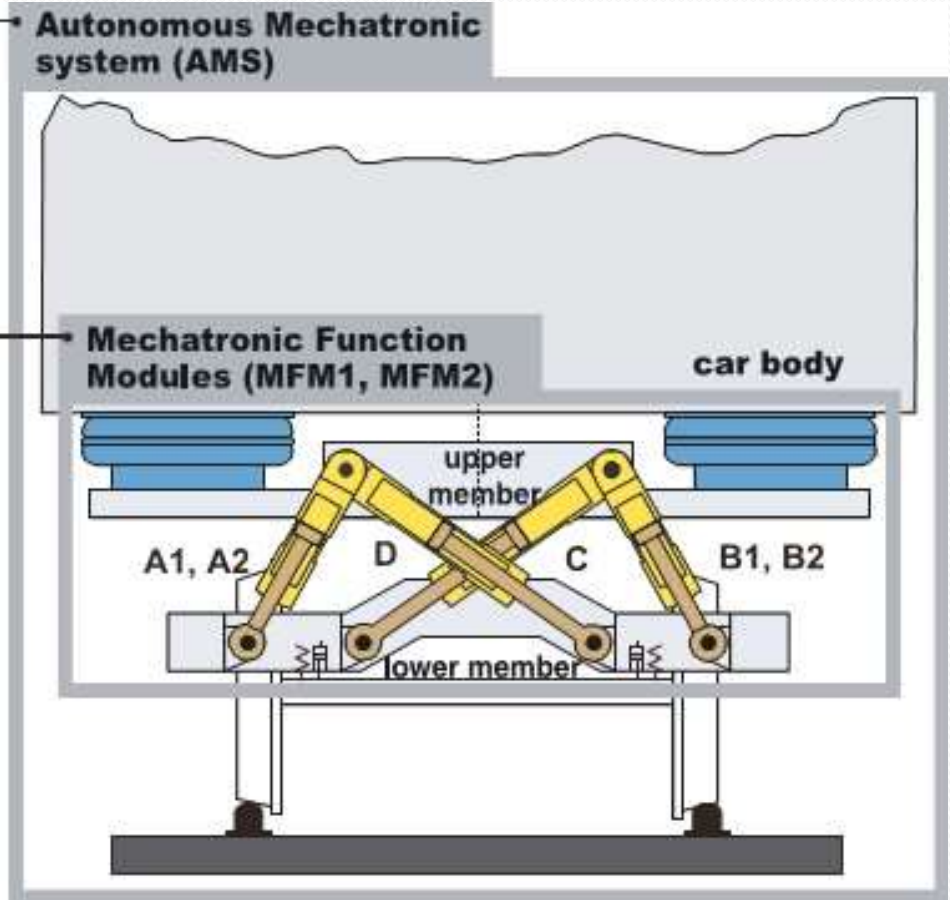


Design of an active spring/tilting module

undertakes superordinate tasks: monitoring the position of the construction (car body)

controlling the individual actors:

- MFM1: controlling the position and speed of the actors
- MFM2: valve activation

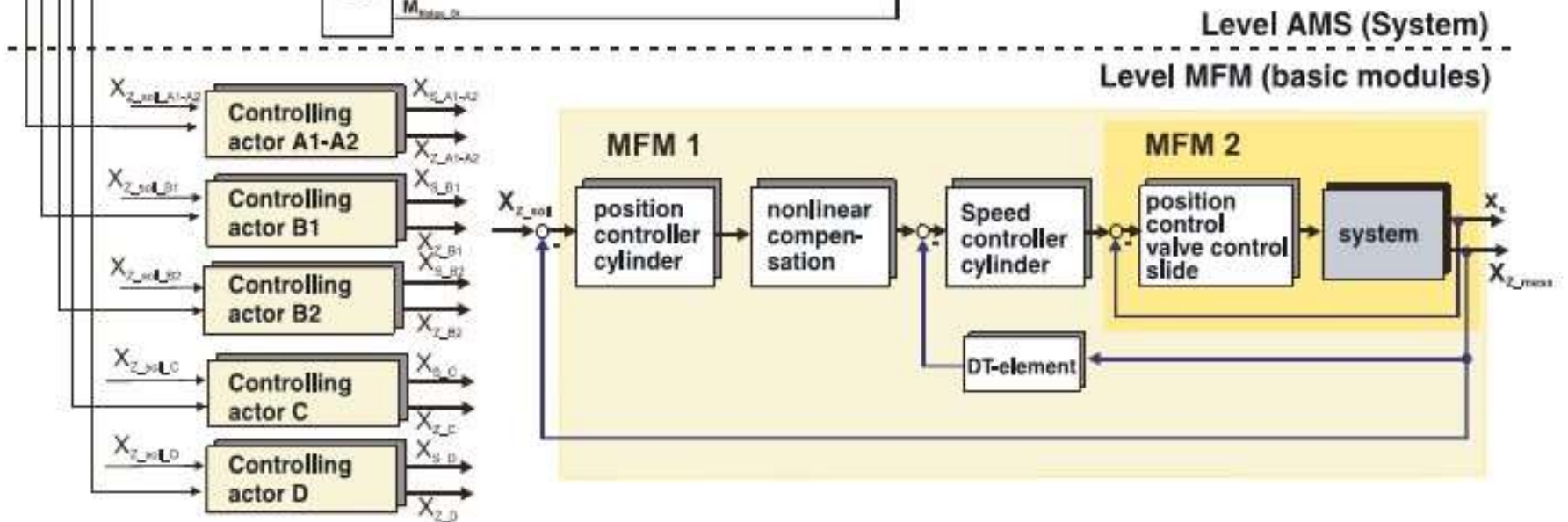
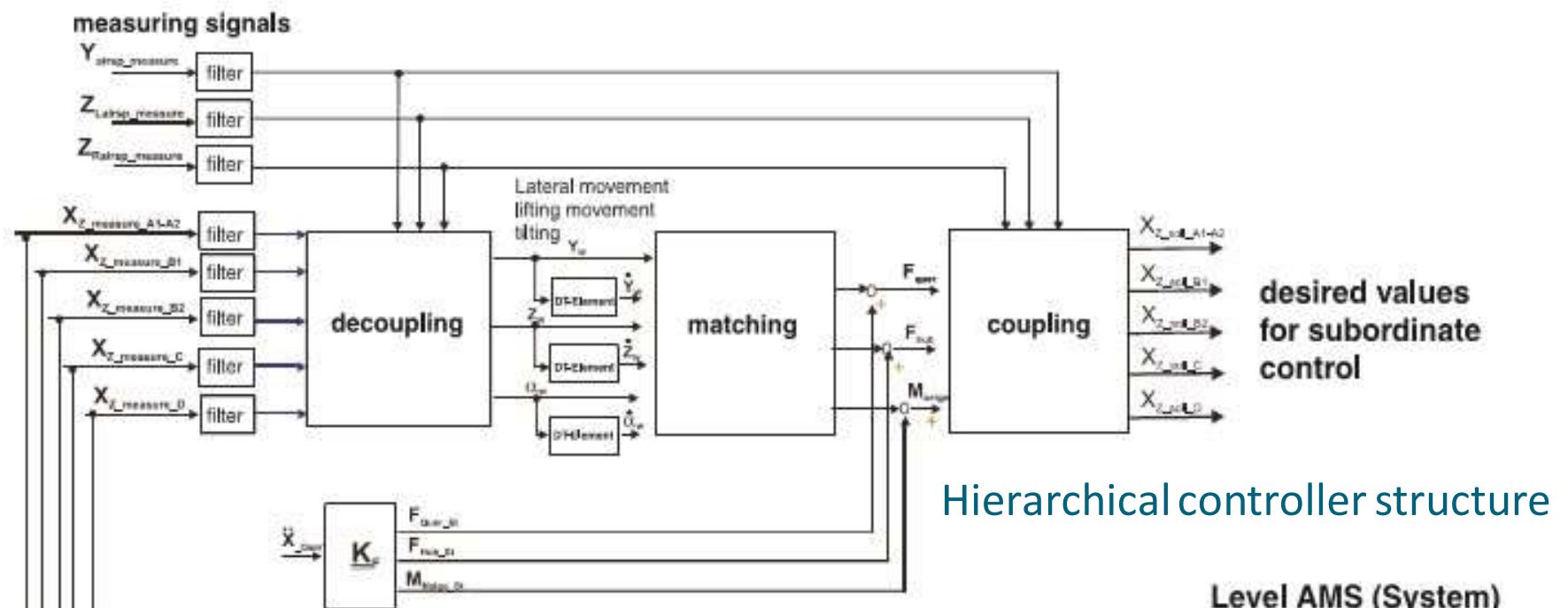


Hierarchization of the overall spring/tilting module and car body system

Design of an active spring/tilting module

❑ Hierarchical system structure

- ❖ the overall system, comprising the spring-tilting module and the car body, can be divided into **two hierarchies**:
 - **On the AMS level**, essentially the position of the construction is monitored,
 - while on the **MFM level** the behavior of the individual actors is considered.
- ❖ The subordinate level can be divided once again into two further levels.
 - MFM1 contains the position and speed control of the hydraulic actors.
 - The activation of a valve takes place via a subordinate MFM (MFM2), which controls the position of the valve slide.



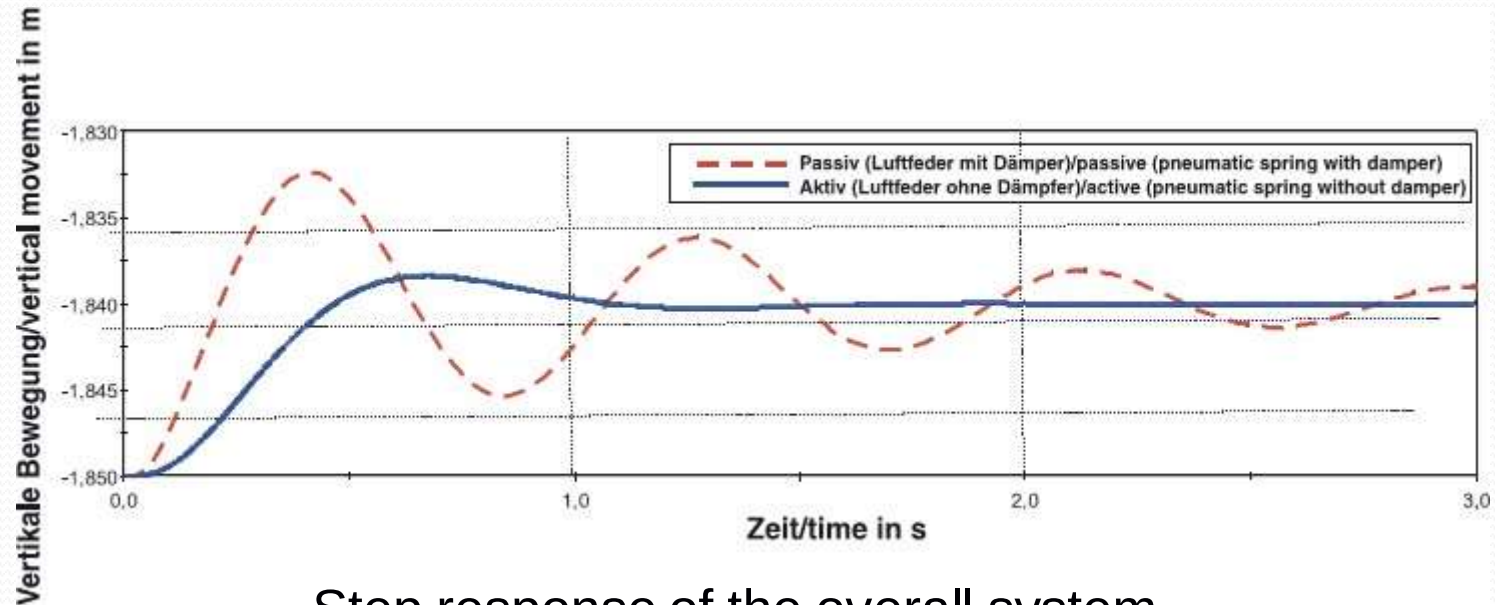
Legende:

$Y_{spring_measure}$, $Z_{Lspring_measure}$, $Z_{Rspring_measure}$
 $X_{Z_measure_XX}$, X_{Z_XX} , X_{Z_meas}
 Y_w , Z_w , $\dot{Y}_w$, $\dot{Z}_w$, α_w , $\dot{\alpha}_w$
 F_{quer_St} , F_{hub_St} , M_{neige_St}

: Displacement of pneumatic springs
 : measured cylinder positions
 : relativ position of track - body
 : disturbing lateral force, lifting force, tilting moment

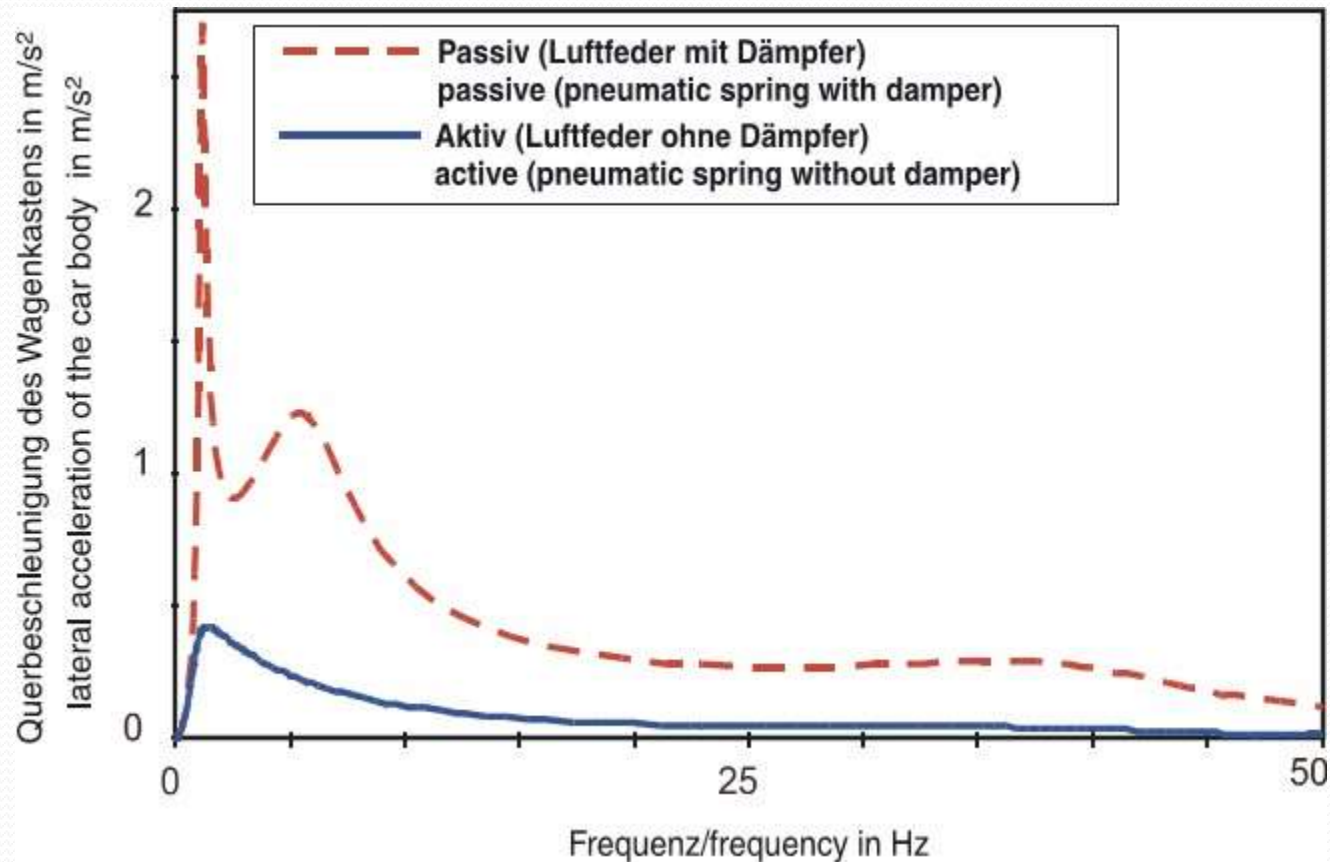
X_{S_XX} : position of valve slides
 $\ddot{x}_{quer}$: lateral acceleration
 X_{Z_XX} , X_{Z_soil} : desired position of cylinders
 F_{quer} , F_{hub} , M_{neige} : transverse force, lifting force, tilting moment

Analysis of the controlled system



Step response of the overall system

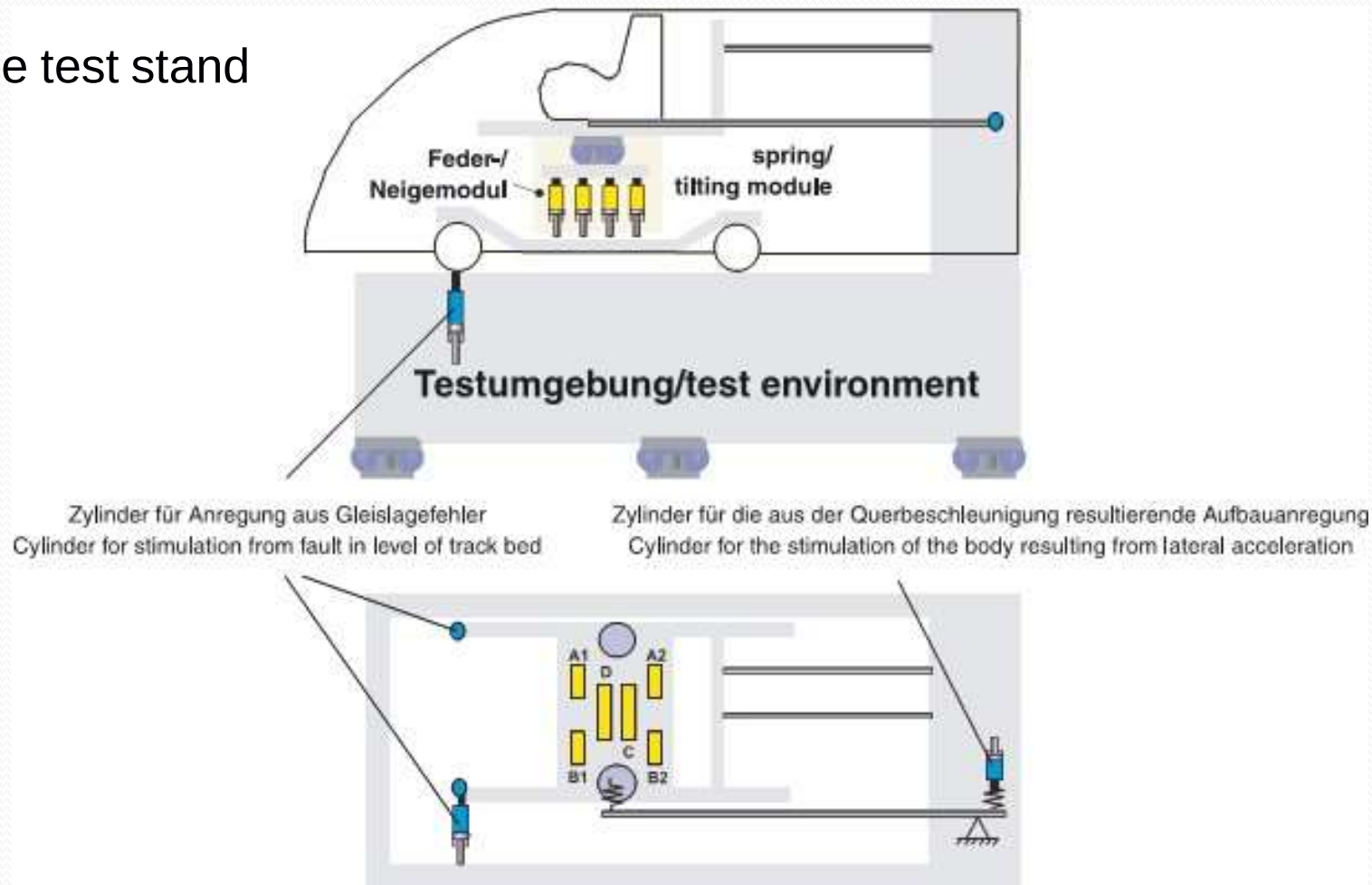
Analysis of the controlled system



Amplitude spectrum of the lateral acceleration

Laboratory trials

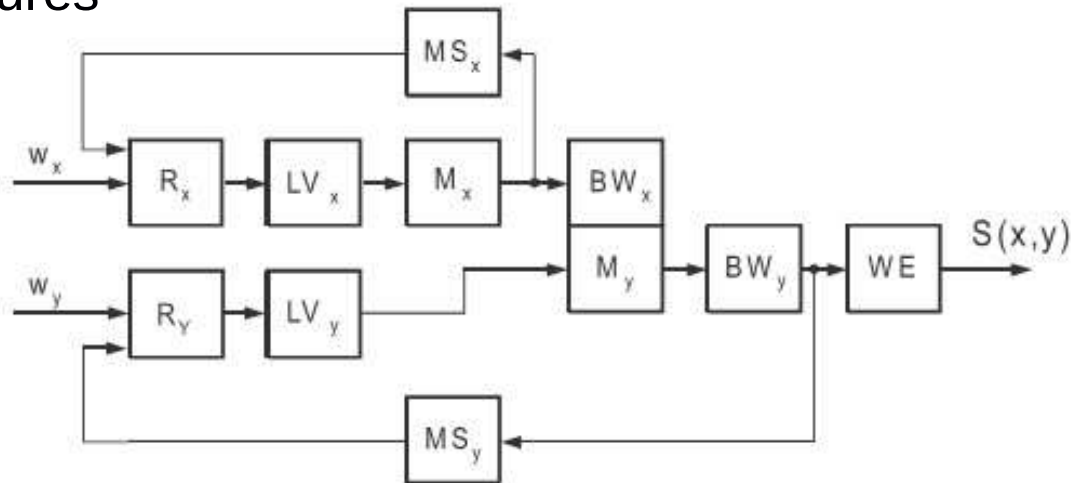
Concept of the test stand



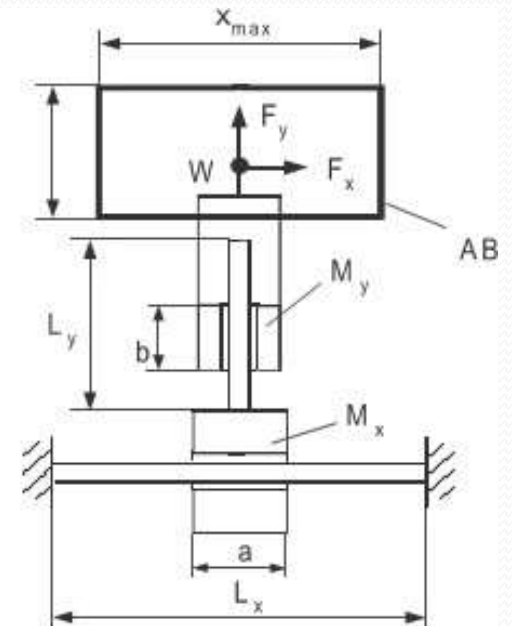
module on a
scale of 1 : 5

Design of integrated

Structures



R	Regler/Controller
LV	Leistungsverstärker/power booster
M	Motor/motor
BW	Bewegungswandler/movement converter
WE	Wirkelement/operating element
MS	Messsystem/measuring system



a, b	Motorlänge/length of motor
AB	Arbeitsbereich/working range
F	Kraft/force
L	Läuferlänge/length of rotor
M	Motor/motor
W	Werkzeug/tool

Two-coordinate drive with serial kinematics
(Page 99 in VDI standard)



Thank You!

Any questions?